

Lecture 1: Computational Cameras

Color

Instructor: Katie Bouman

TAs: Brandon Zhao, Brady Bhalla, Zachary Huang

Course announcements

- Office Hour Locations. Thursday, Friday, and Monday in Annenberg 104 . Friday in Katie's office which is Annenberg 346
- Office Hours: Since there is no homework this week we do not plan to hold office hours until Next Thursday. But if you have a question please just email us and we can join office hours or find another time to answer your questions
- Write down your name and what question you answered or asked during class and leave by the door
- Pick up Raspberry Pi materials on Tuesday, so please make sure to come on time, or a bit early

Overview of today's lecture

- Introduction to color
- Human vision
- Color spaces

Slide credits

Many of these slides were borrowed/adapted from:

- Ioannis (Yannis) Gkioulekas (CMU)
- John Austin (<https://johnaustin.io/>)
- Mark Levoy

What is Color?

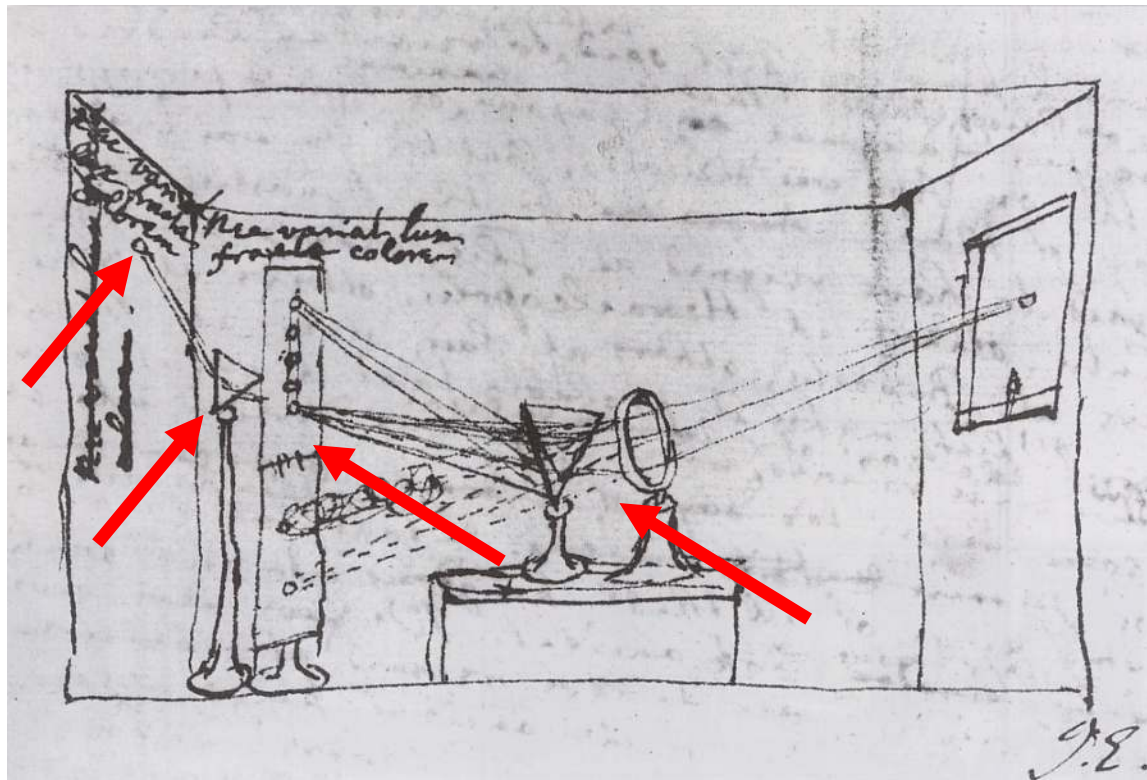
an experience!

caused by our sensitivity to different wavelengths of light

Sunlight can be divided only once



Isaac Newton
(1643-1727)

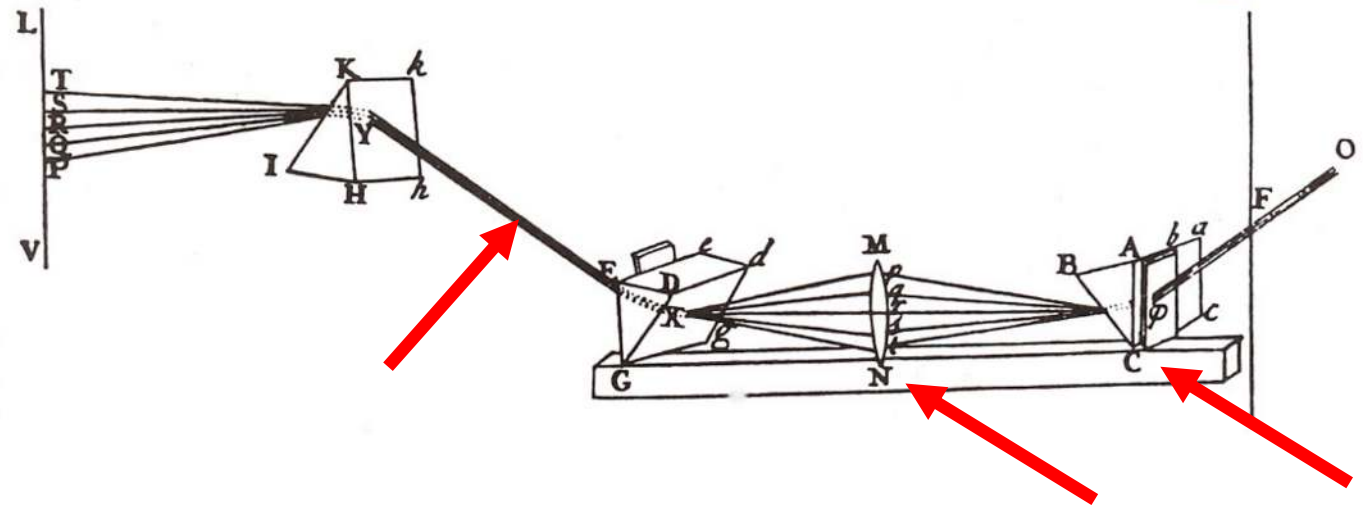


experiment performed 1665, drawing made in 1672

Sunlight can be divided only once

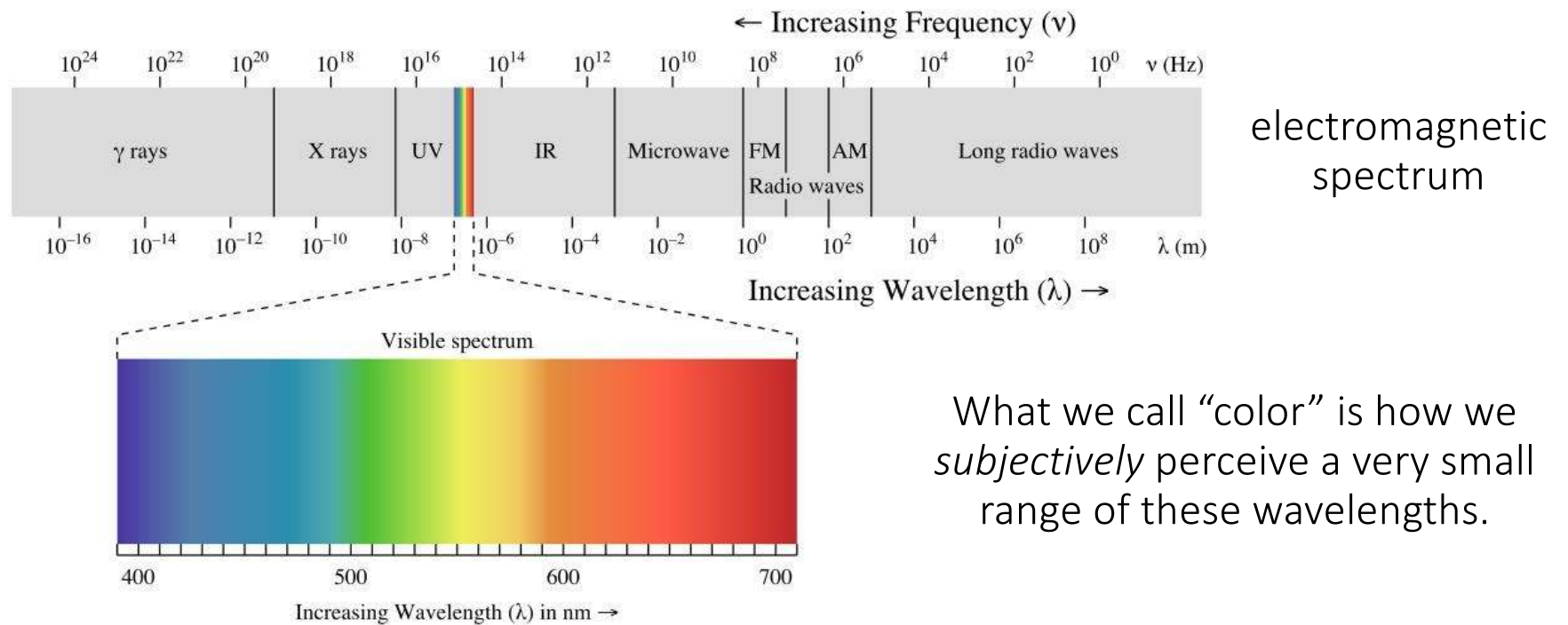


Isaac Newton
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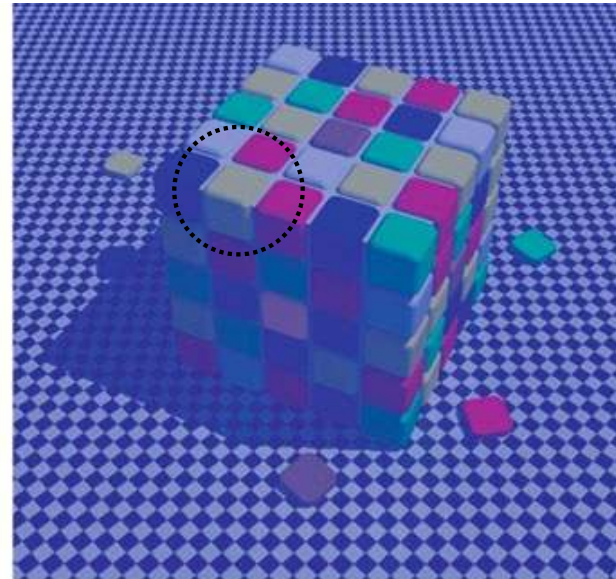
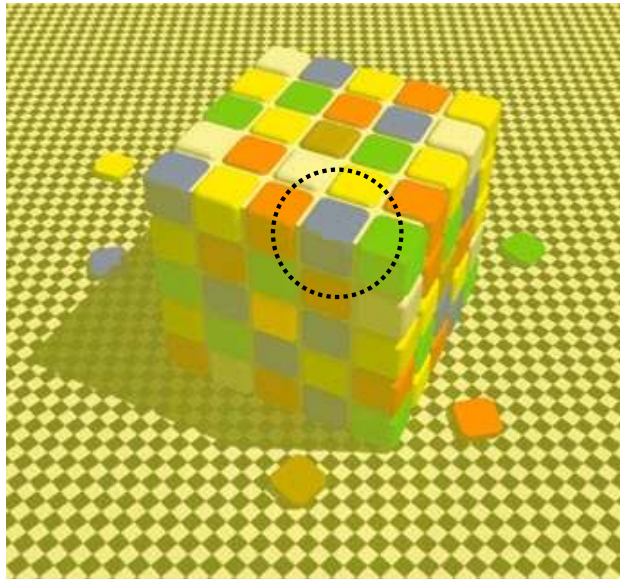


Color is an artifact of human perception

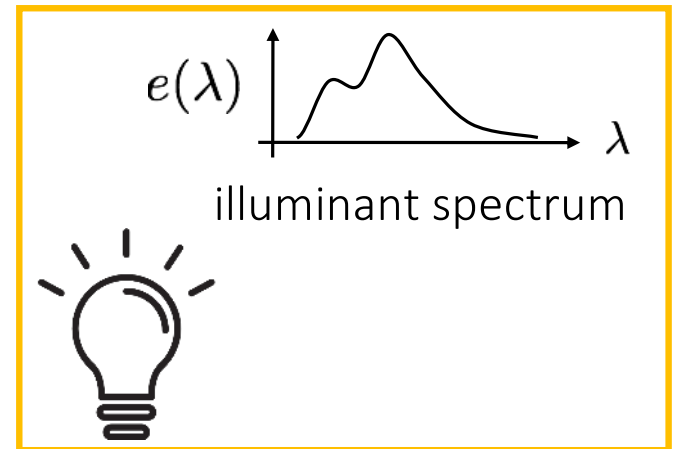
- “Color” is not an *objective* physical property of light (electromagnetic radiation).
- Instead, light is characterized by its wavelength.



Color is subjective



Light-material interaction

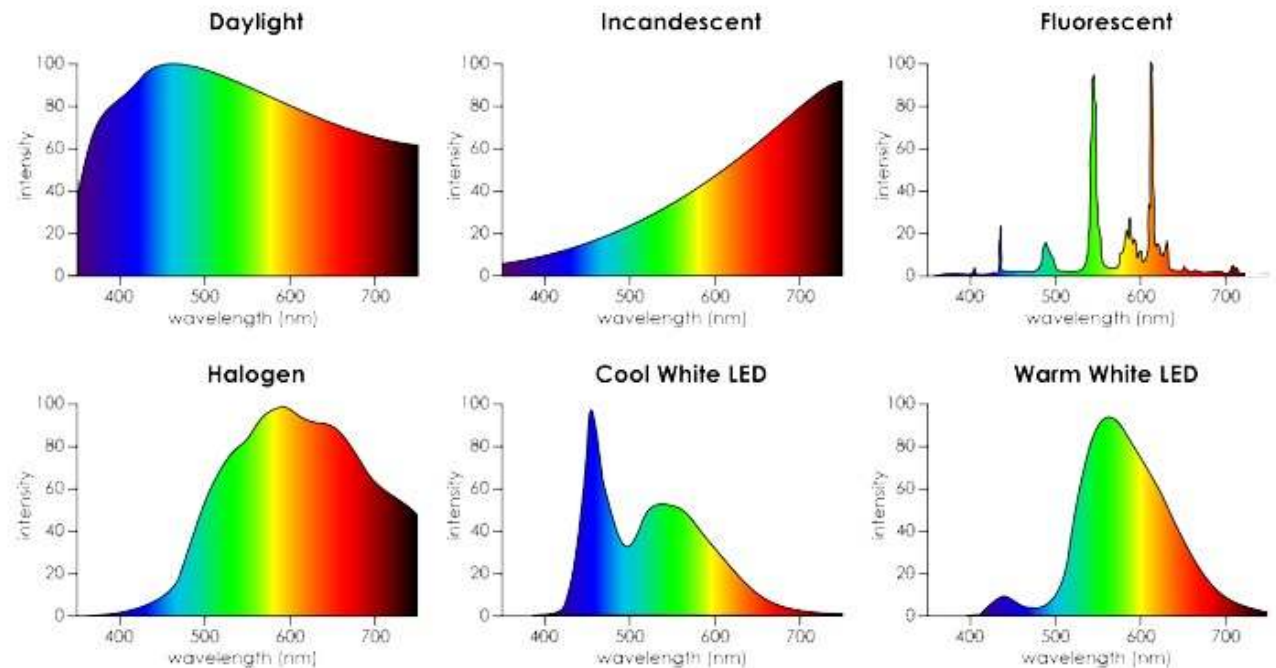


Illuminant Spectral Power Distribution (SPD)

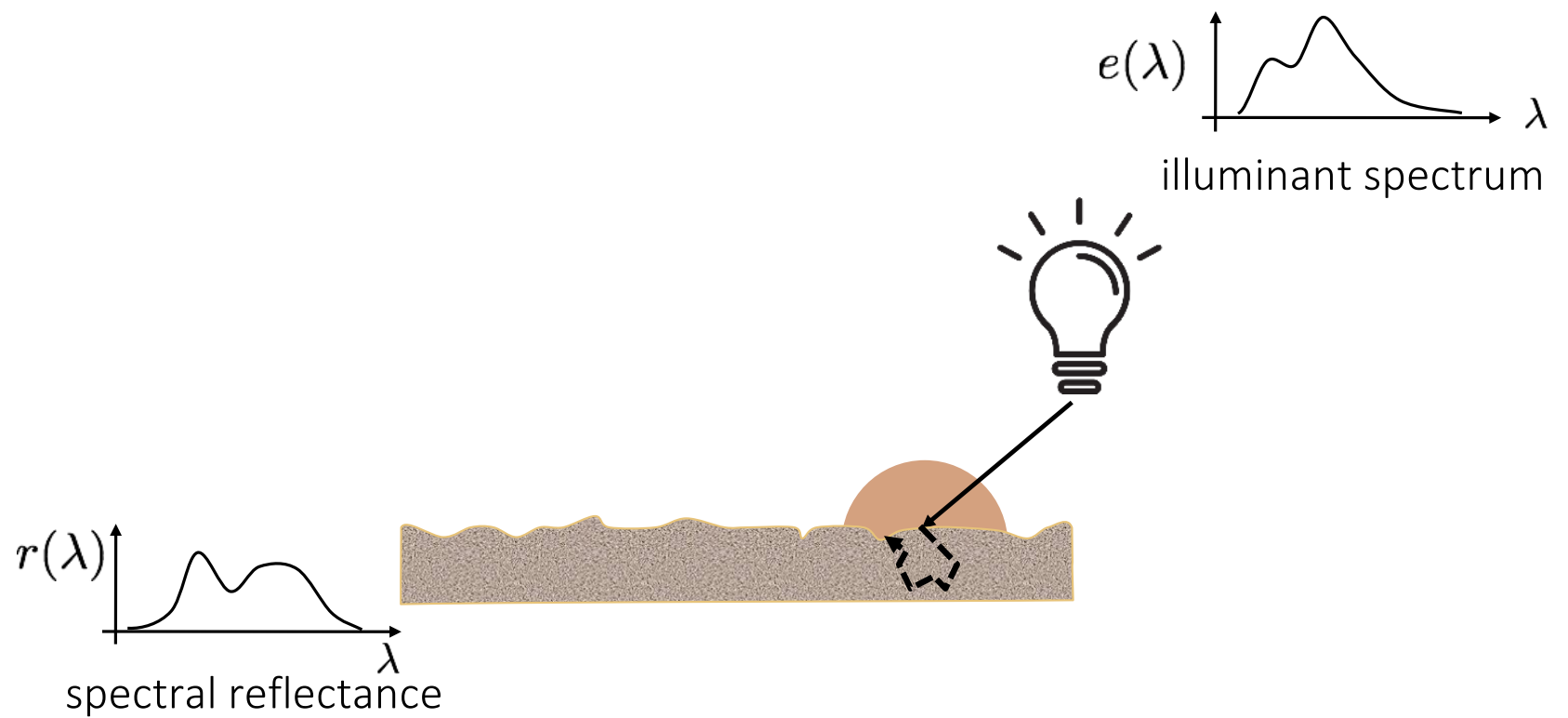
- Most types of light “contain” more than one wavelengths.
- We can describe light based on the distribution of power over different wavelengths.



We call our sensation of all of these distributions “white”.

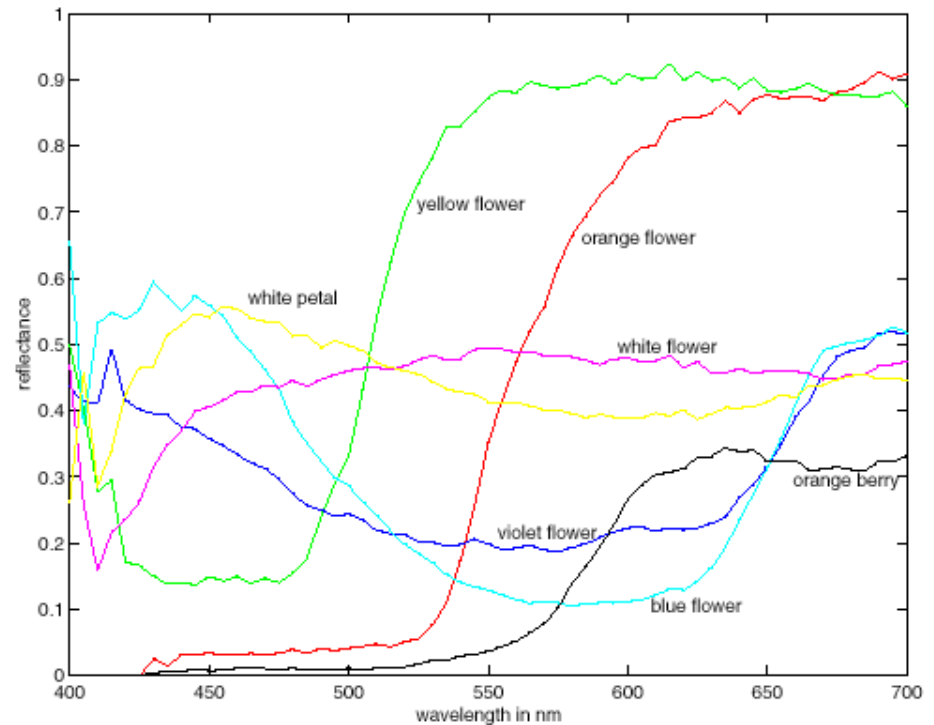


Light-material interaction

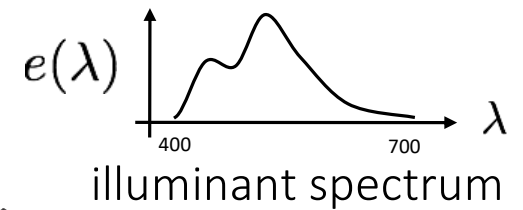
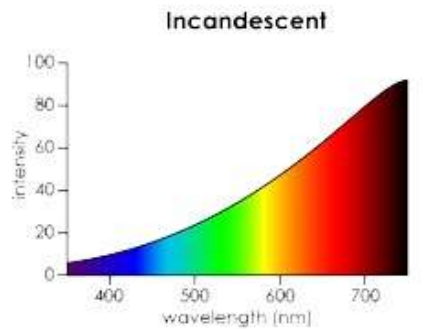
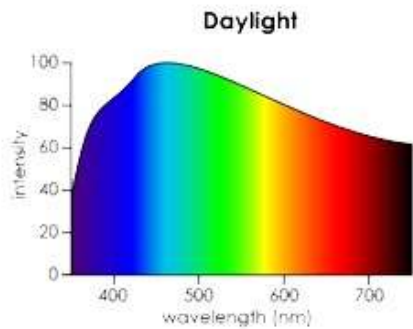


Spectral reflectance

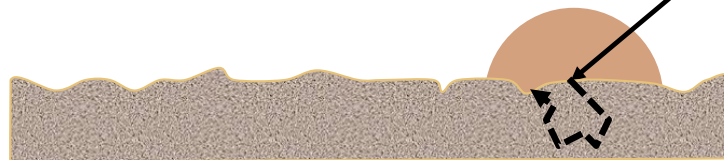
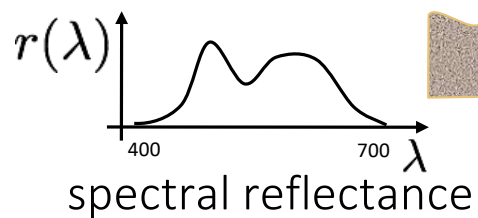
- Most materials absorb and reflect light differently at different wavelengths.
- We can describe this as a ratio of reflected vs incident light over different wavelengths.



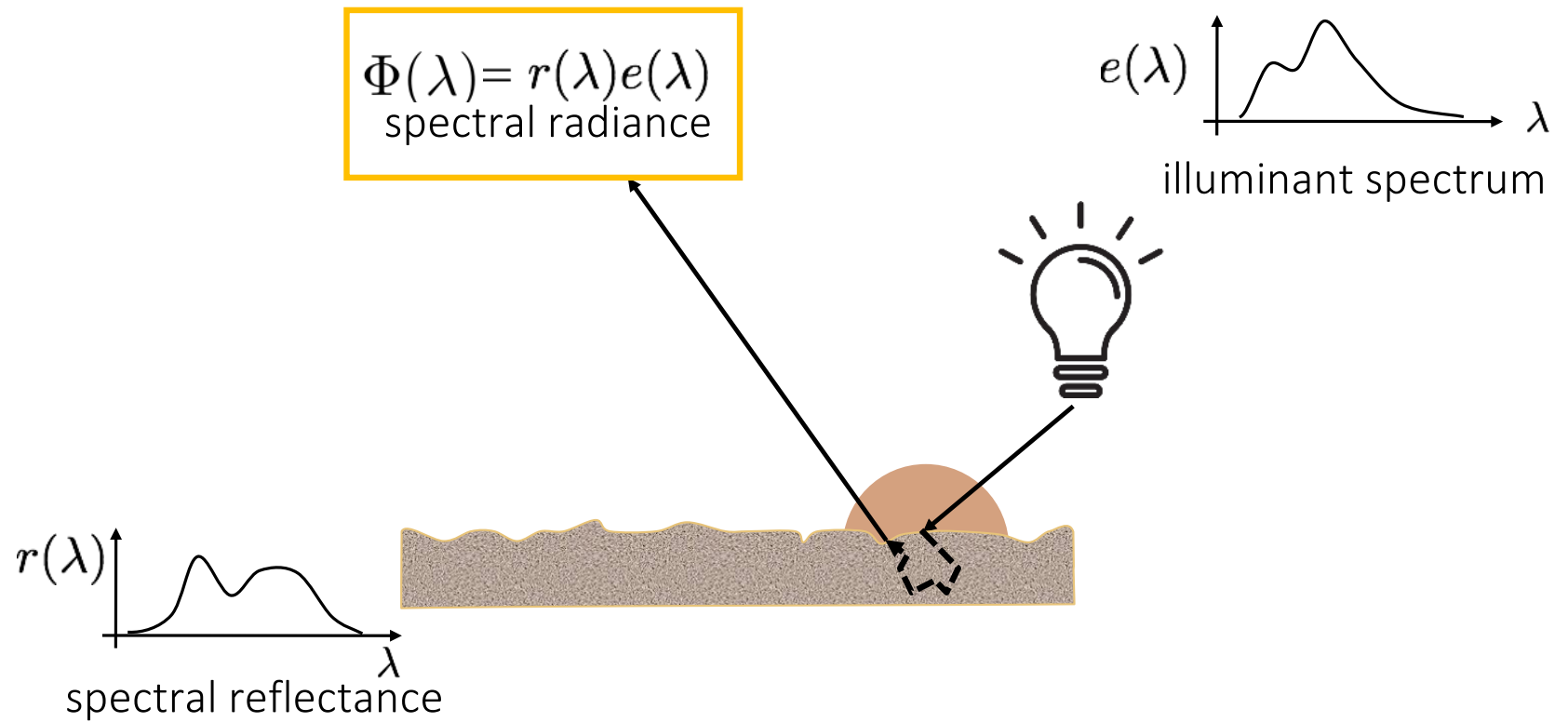
Light-material interaction



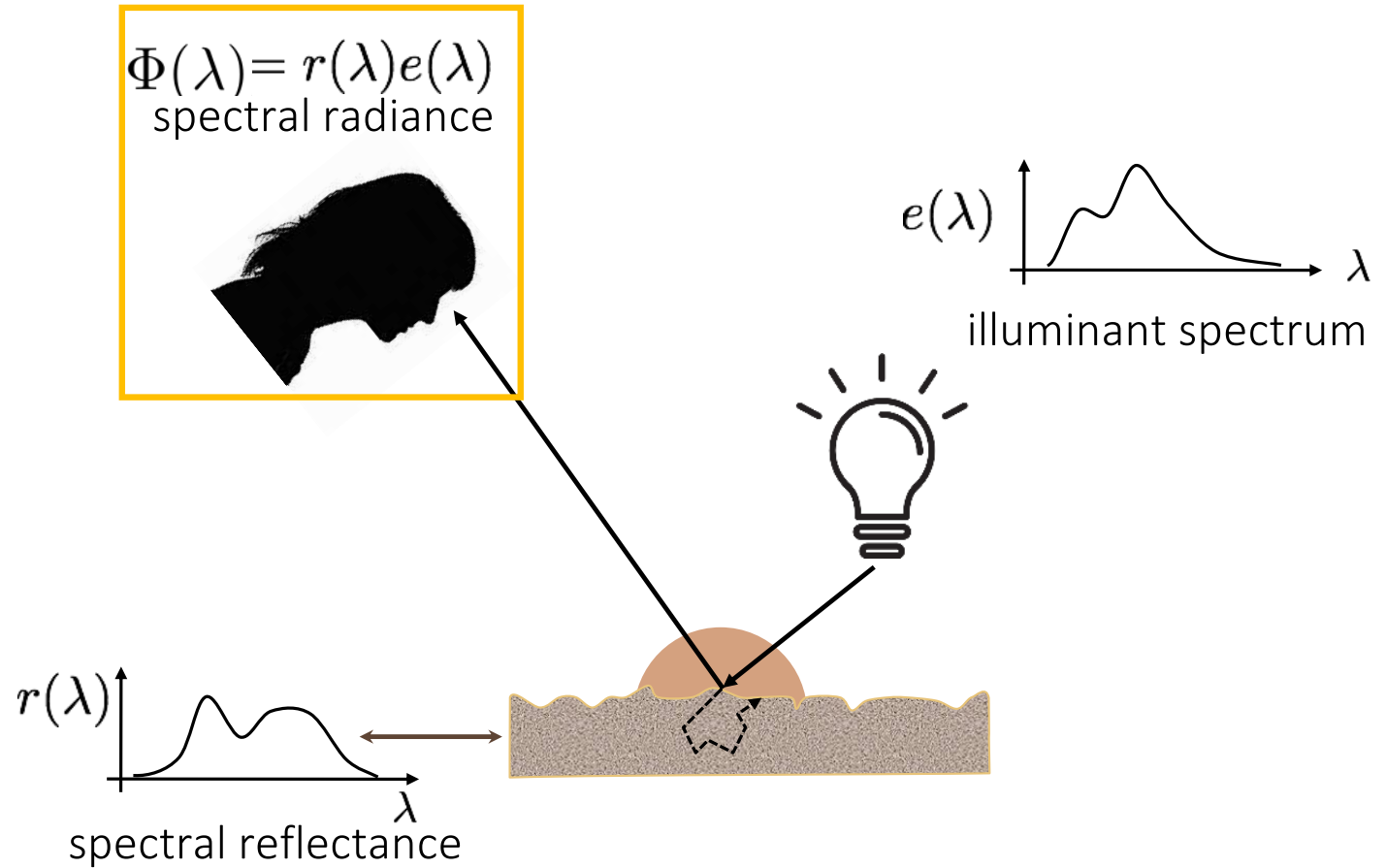
What color do you think this material would look like?



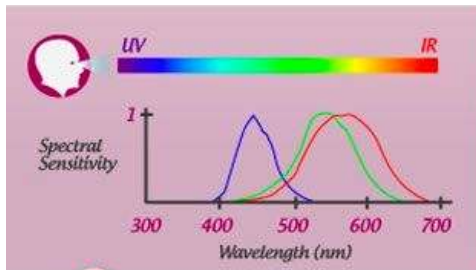
Light-material interaction



Light-material interaction

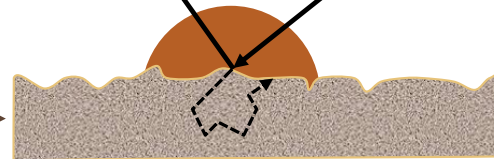
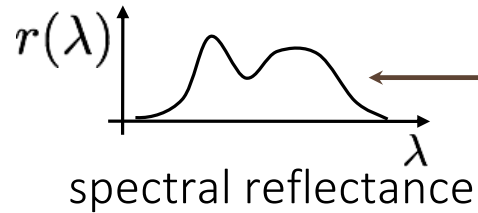
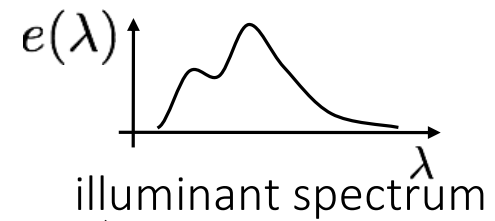
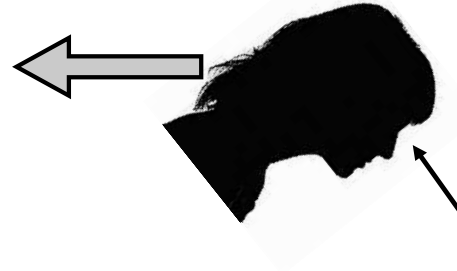


Human color vision



$$\Phi(\lambda) = r(\lambda)e(\lambda)$$

spectral radiance



Spectral Sensitivity Function (SSF)

- When measuring light of $\Phi(\lambda)$, a sensor produces a *scalar* response according to the sensor's *spectral sensitivity function* $f(\lambda)$

sensor response $\longrightarrow R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$

light $\downarrow$ sensor SSF $\downarrow$

Weighted combination of light's SPD: light contributes more at wavelengths where the sensor has higher sensitivity.

Monochromats

Australian Sea Lion

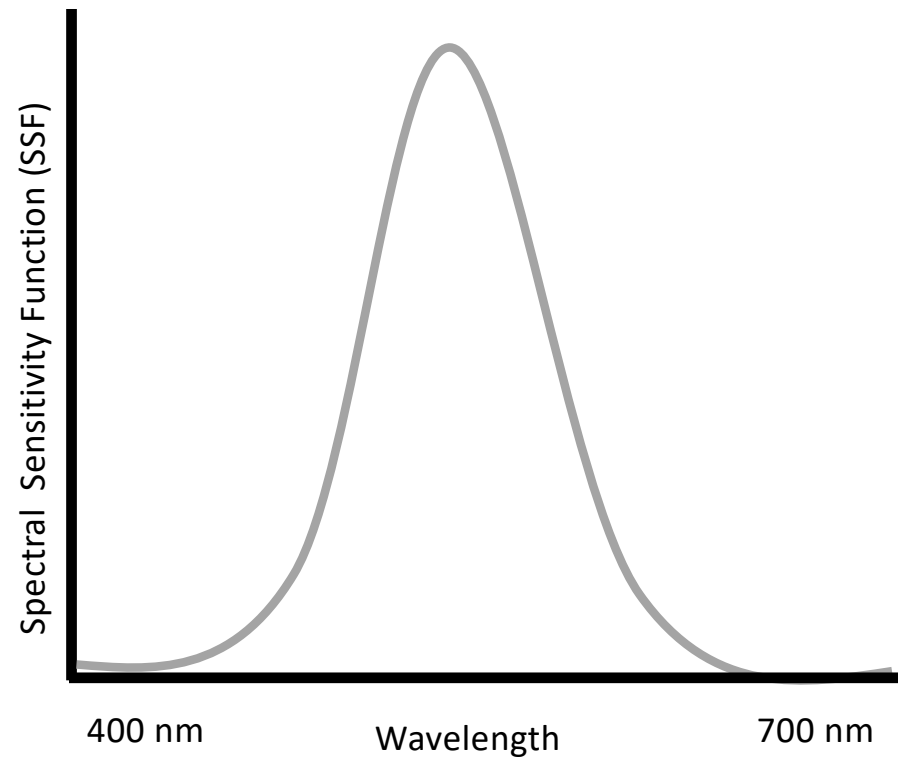


Can it see different colors?

response
↓
 $R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$

light
↓
 $\Phi(\lambda)$

SSF
↓
 $f(\lambda)$



Monochromats

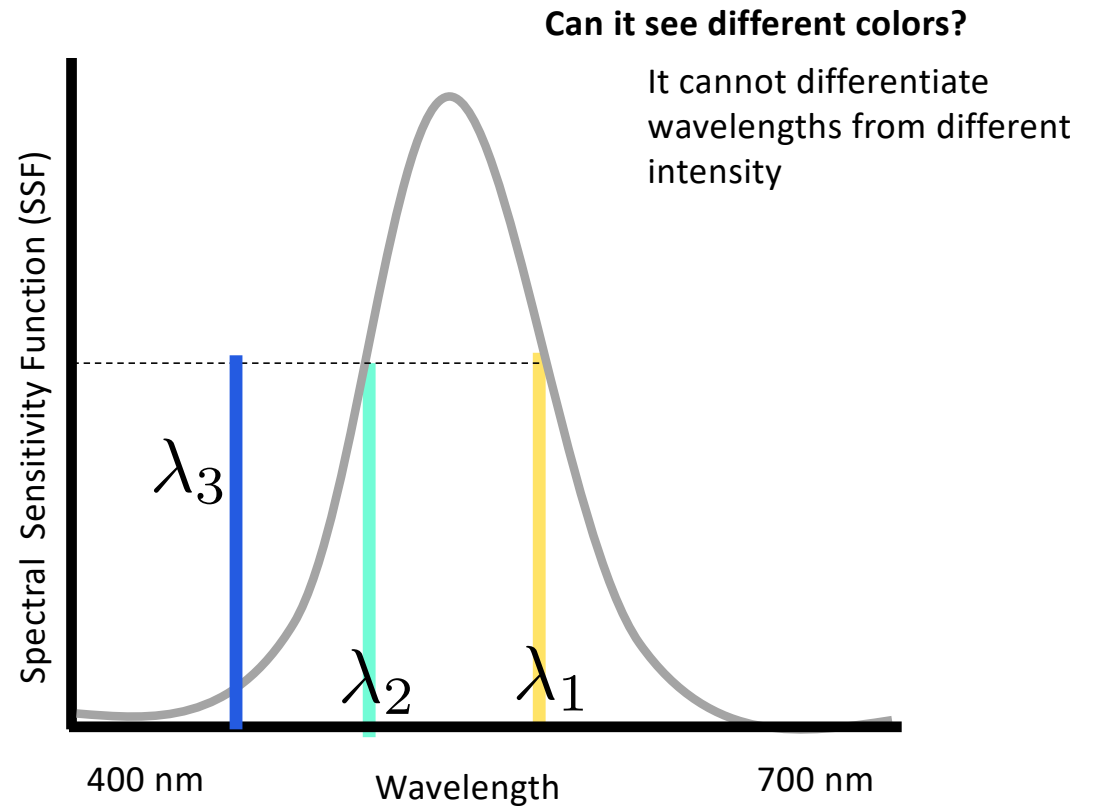
Australian Sea Lion



response
↓
$$R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$$

light
↓
 $\Phi(\lambda)$

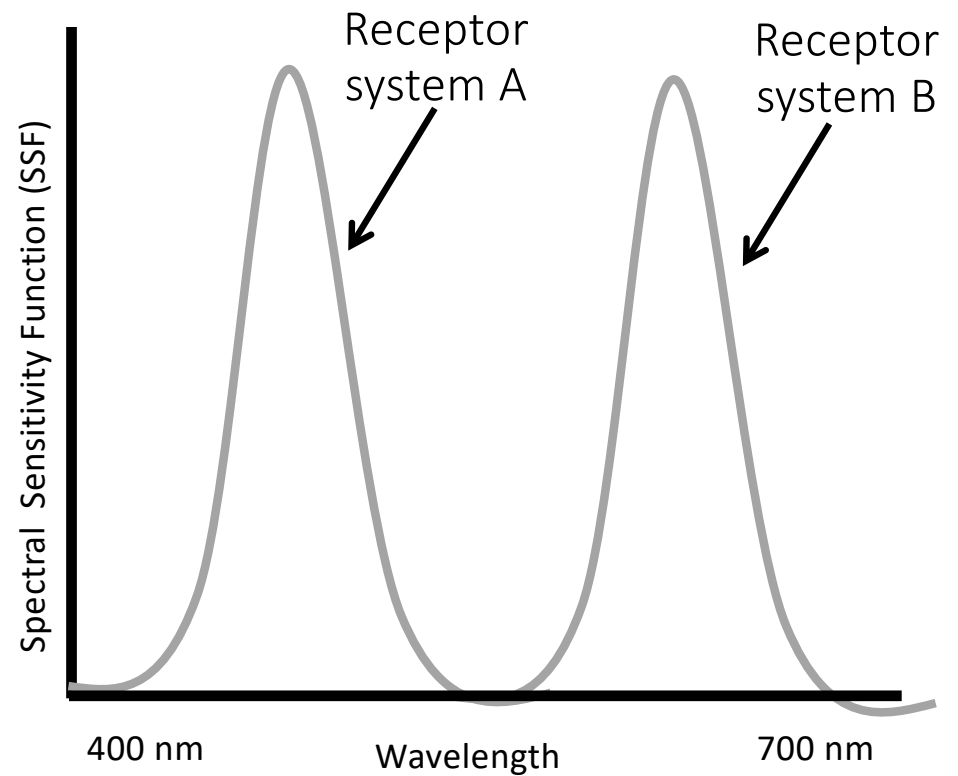
SSF
↓
 $f(\lambda)$



Dog



Dichromats



response
↓
 $R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$

light
↓
 $\Phi(\lambda)$

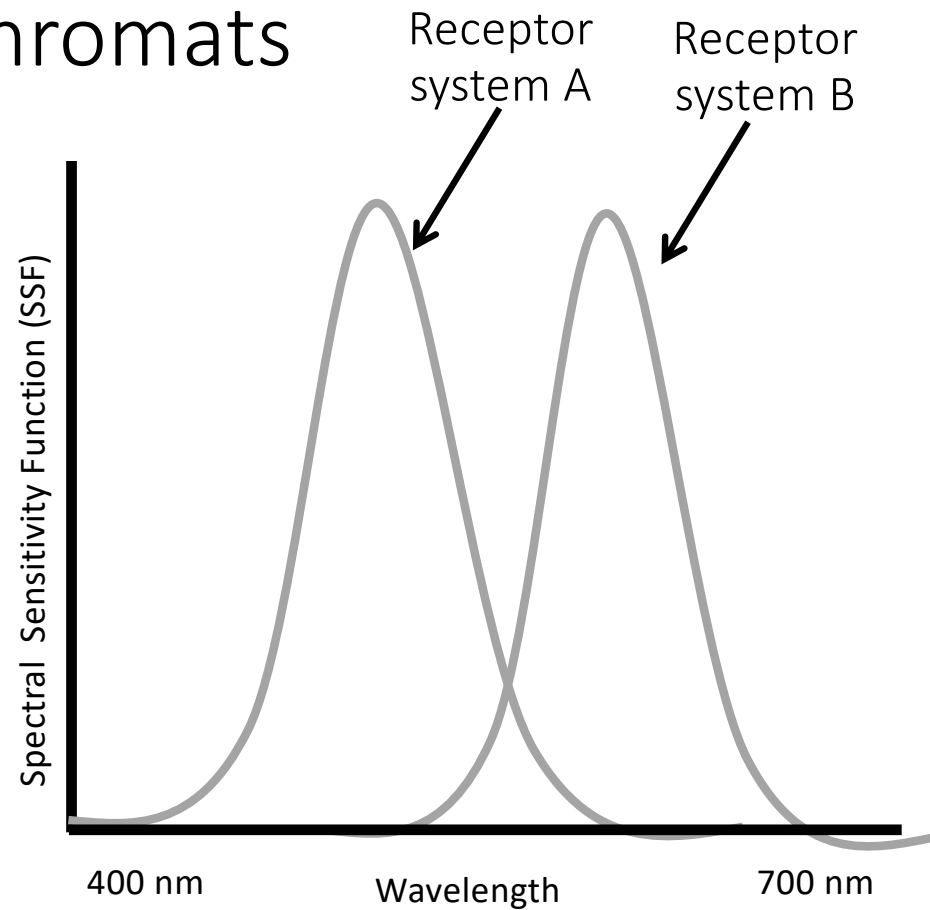
SSF
↓
 $f(\lambda)$

This dog can see 2 colors. How do we design a better dog?

Dog



Dichromats



response

light

SSF

$$R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$$

How many colors can
this “dog” see?

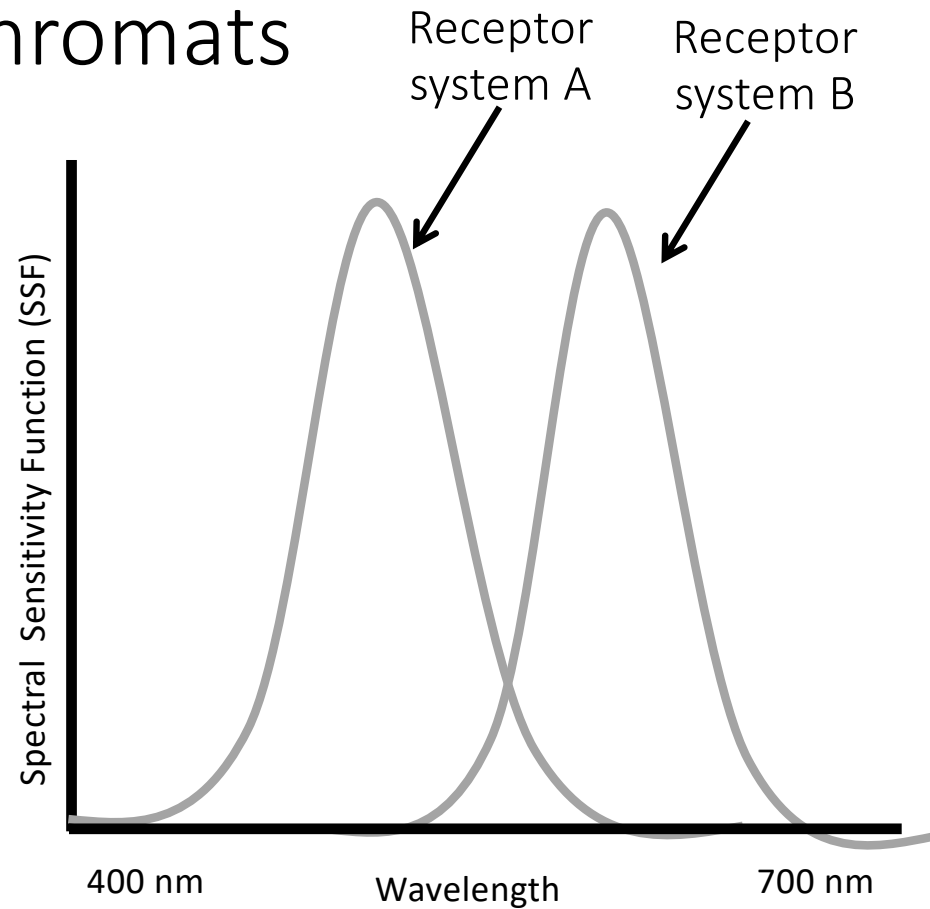


3



Infinite

Dichromats



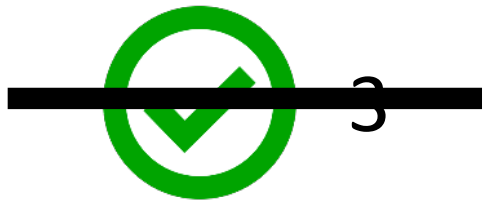
response

light

SSF

$$R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$$

How many colors can
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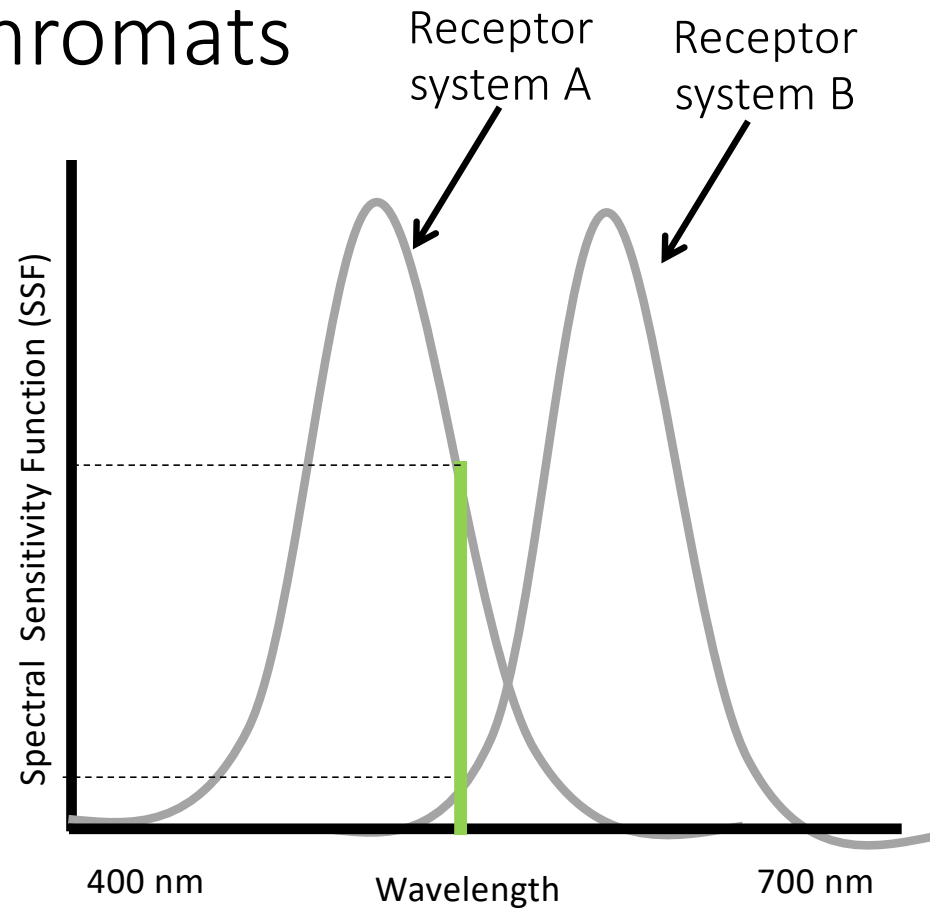


3



Infinite

Dichromats



response

light

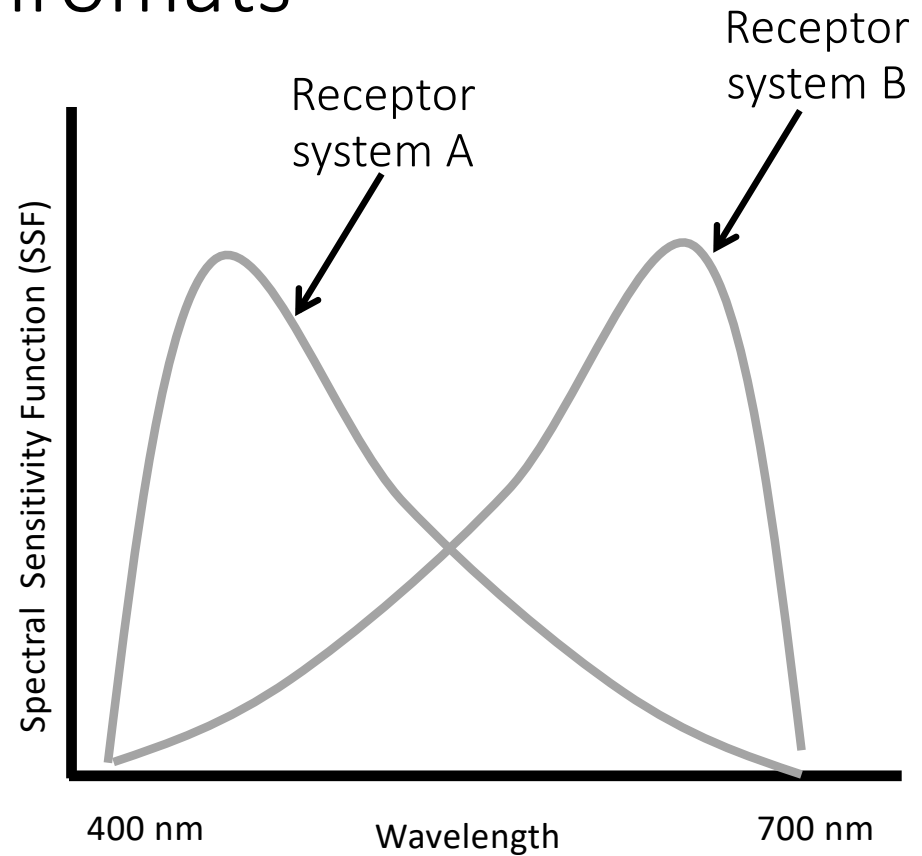
SSF

$$R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$$

Dog



Dichromats



response
↓
 $R = \int_{\lambda} \Phi(\lambda) f(\lambda) d\lambda$

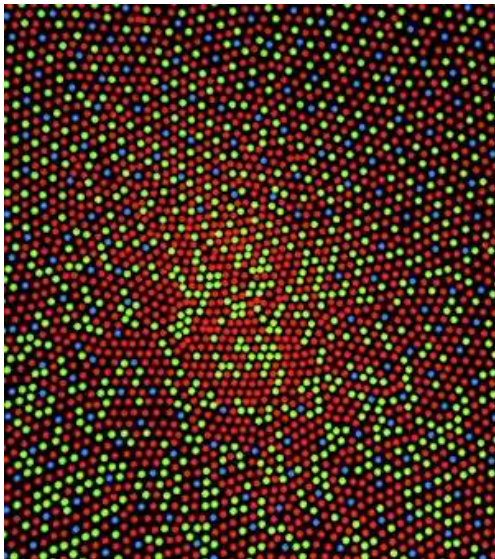
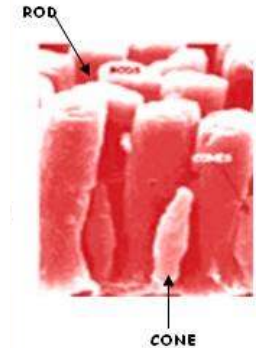
light
↓
 $\Phi(\lambda)$

SSF
↓
 $f(\lambda)$

It now CAN differentiate a change in wavelengths from a change in intensity

Spectral Sensitivity Function of Human Eye

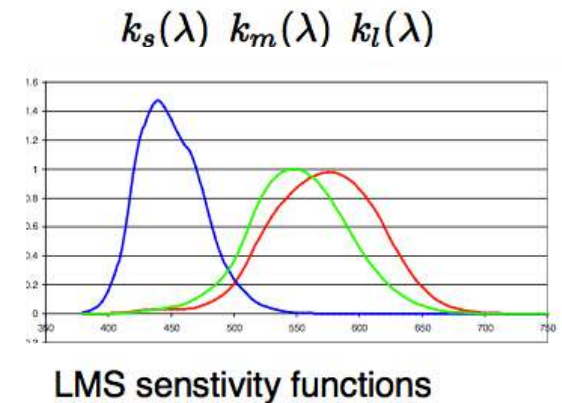
- The human eye is a collection of light sensors called rod & cone cells.
- There are three types of cells with different spectral sensitivity functions.
- Human color perception is three-dimensional (*tristimulus color*).



“short” $S = \int_{\lambda} \Phi(\lambda) S(\lambda) d\lambda$

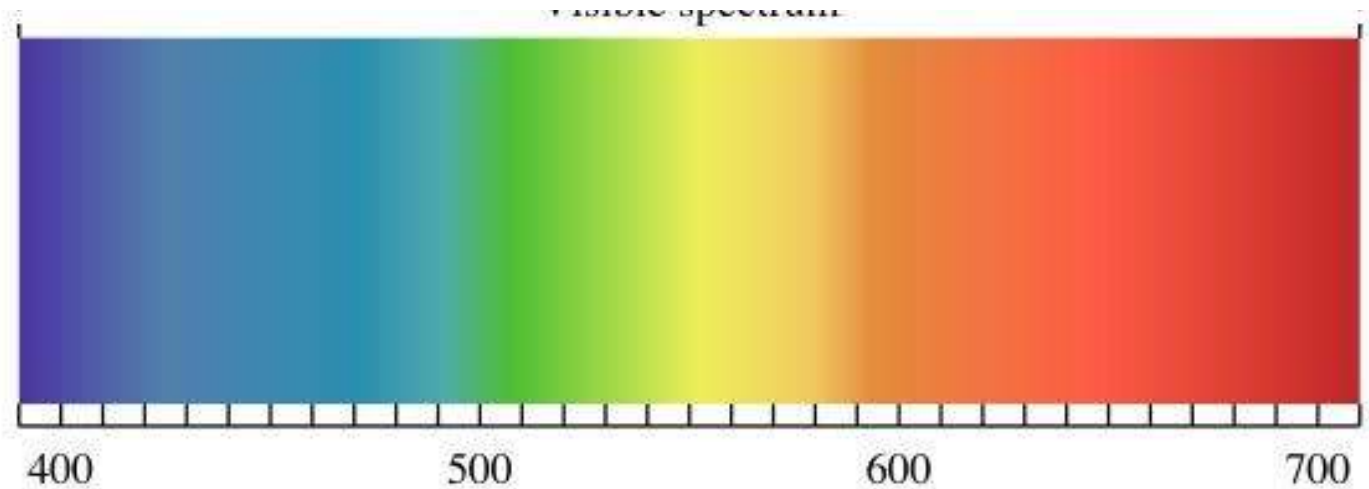
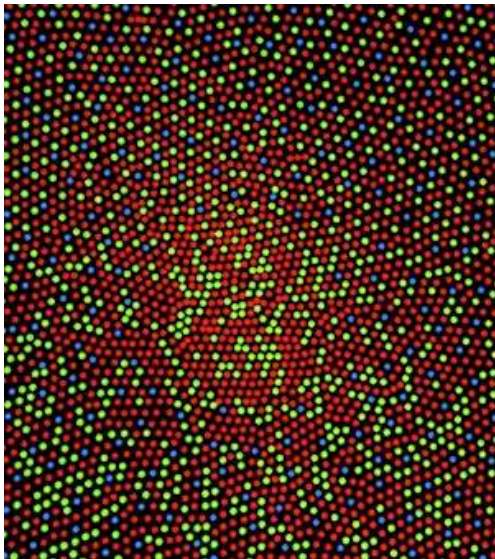
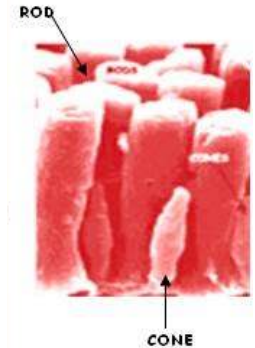
“medium” $M = \int_{\lambda} \Phi(\lambda) M(\lambda) d\lambda$

“long” $L = \int_{\lambda} \Phi(\lambda) L(\lambda) d\lambda$



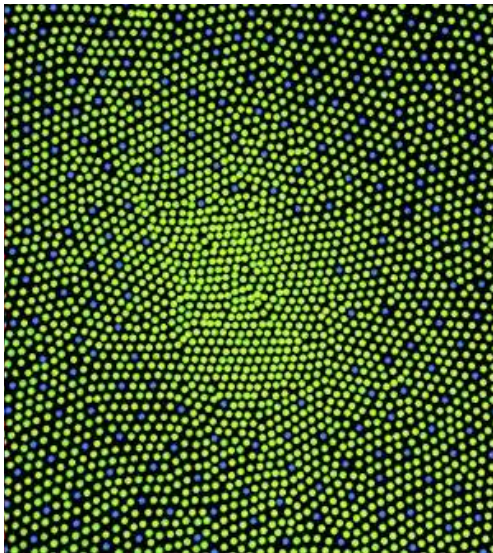
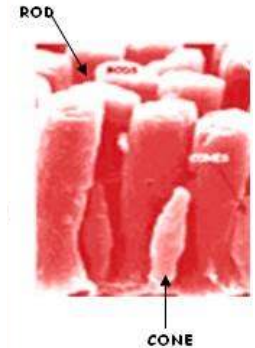
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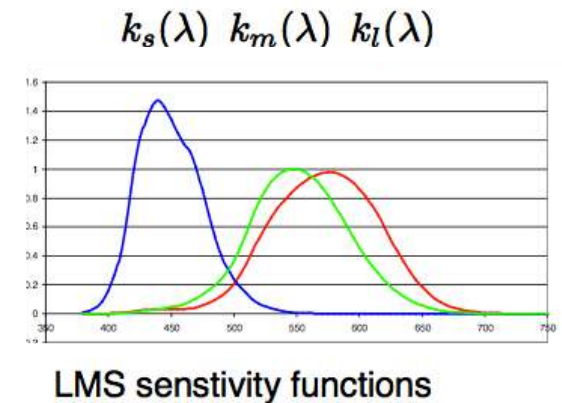
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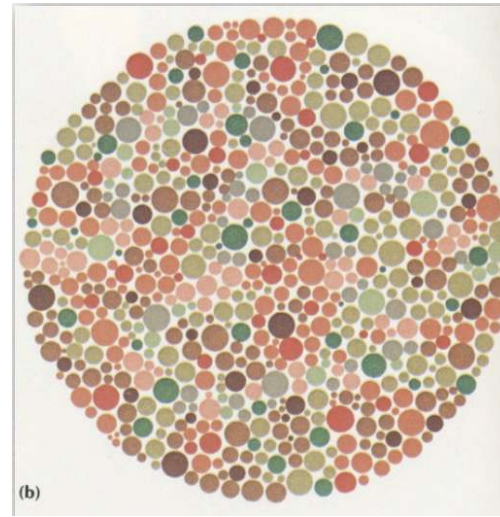
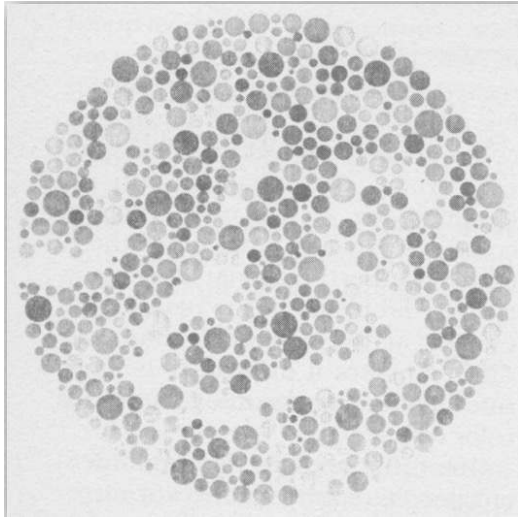
“short” $S = \int_{\lambda} \Phi(\lambda) S(\lambda) d\lambda$

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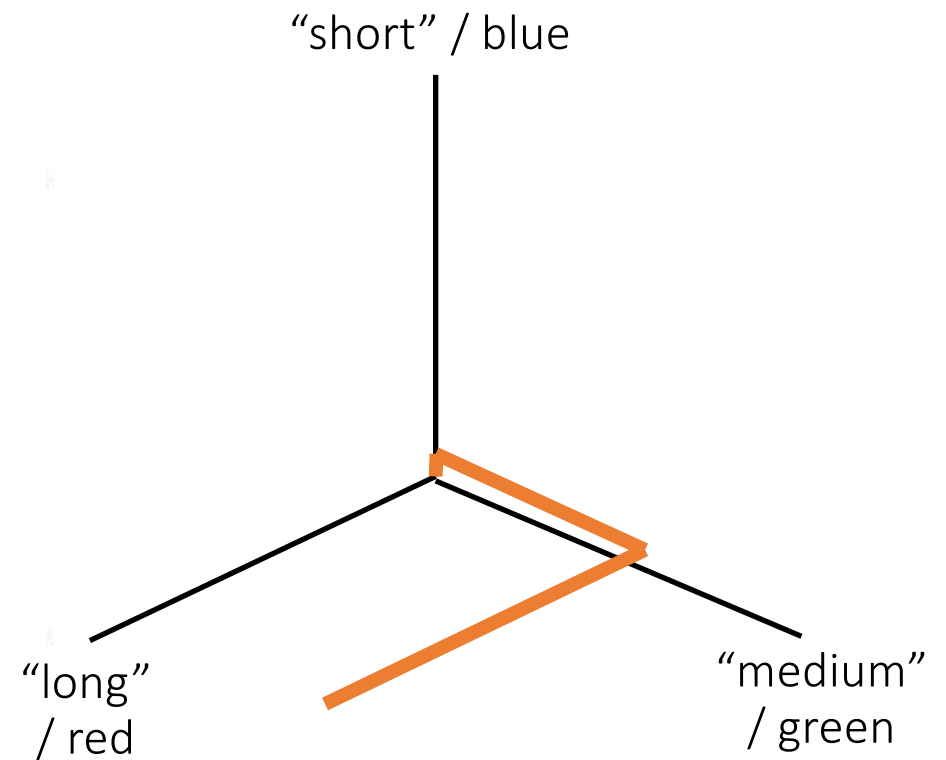
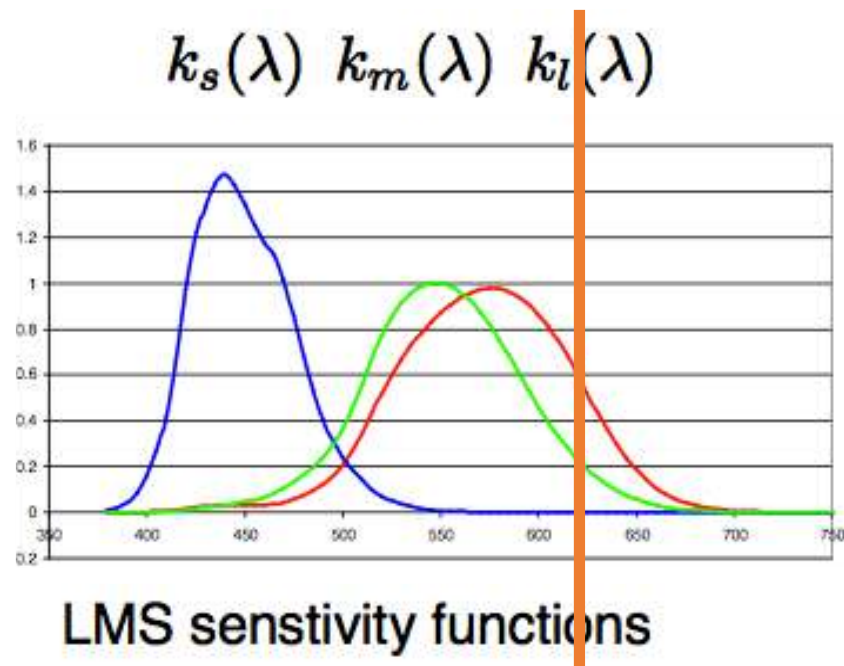


An advantage of being colorblind

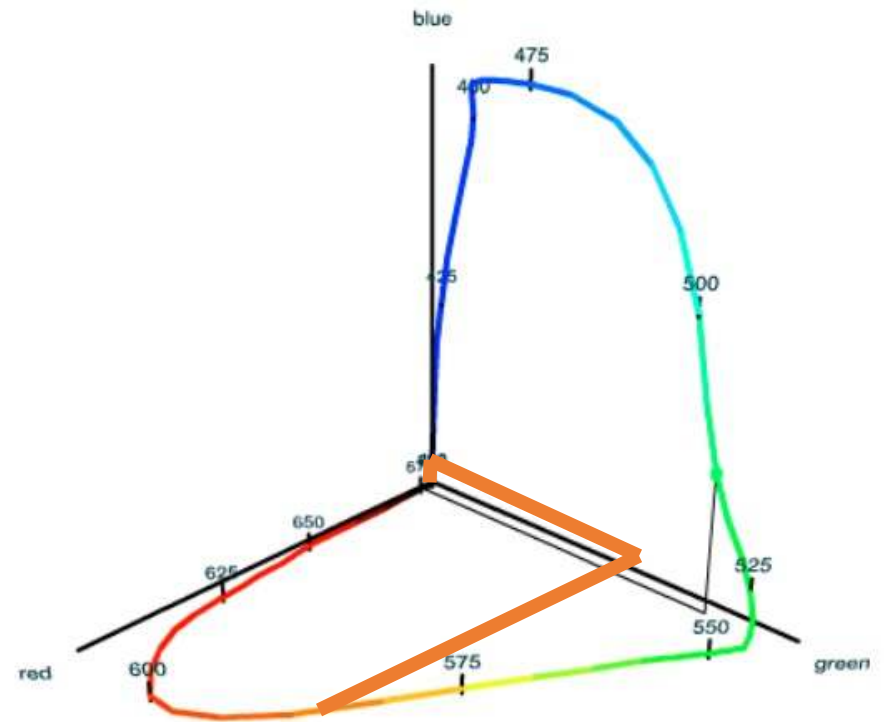
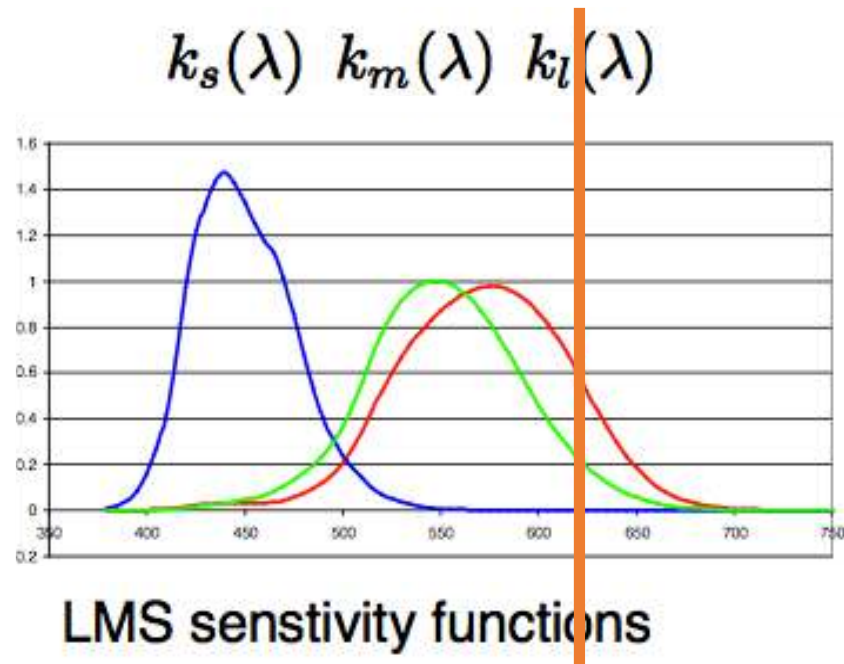


- the maze (at left) is recreated (at right) using subtle intensity differences, but overridden by stronger red-green color differences
- Someone with red-green colorblindness can much more easily see the maze on the right

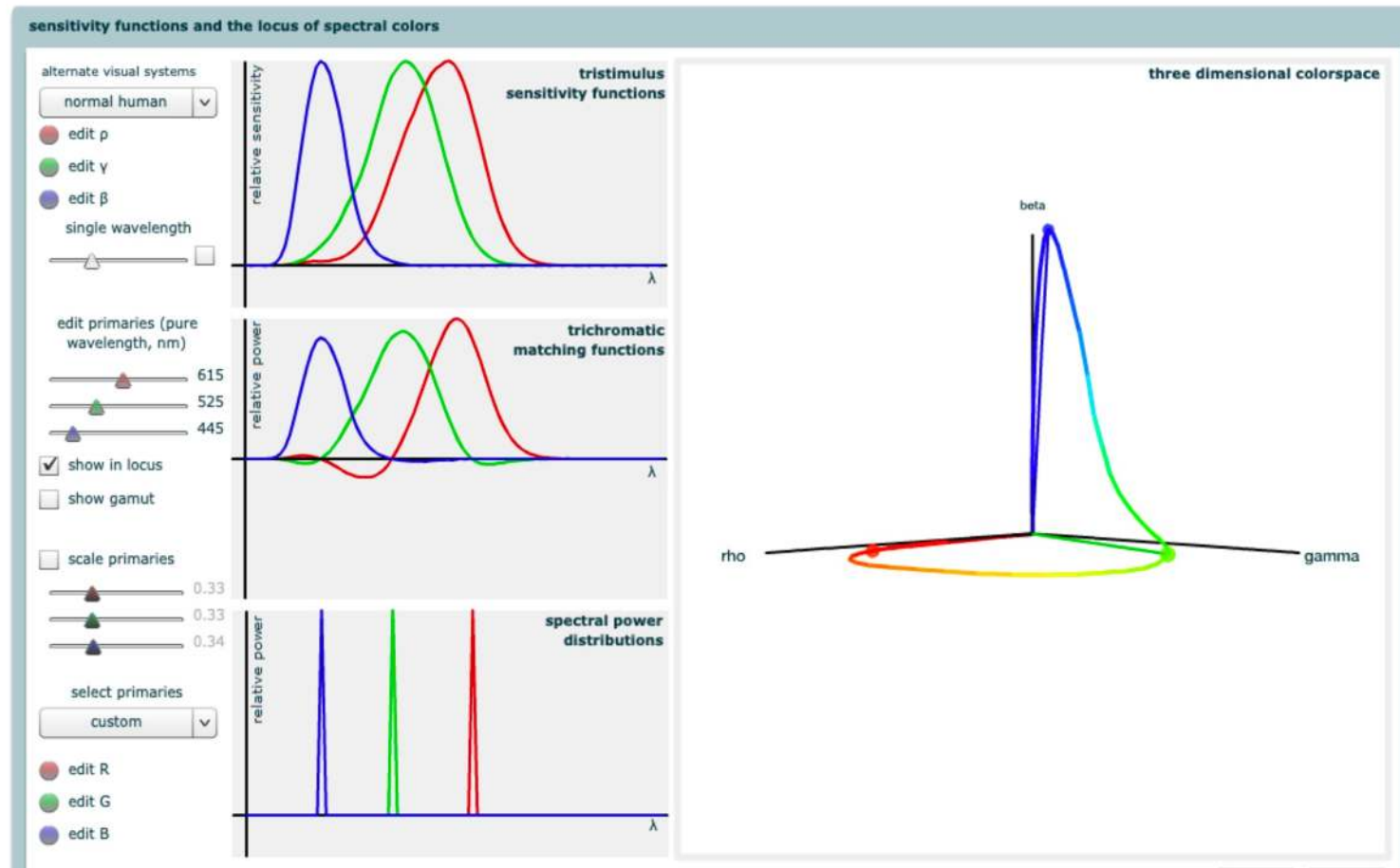
Locus of Spectral Colors / Spectral Locus



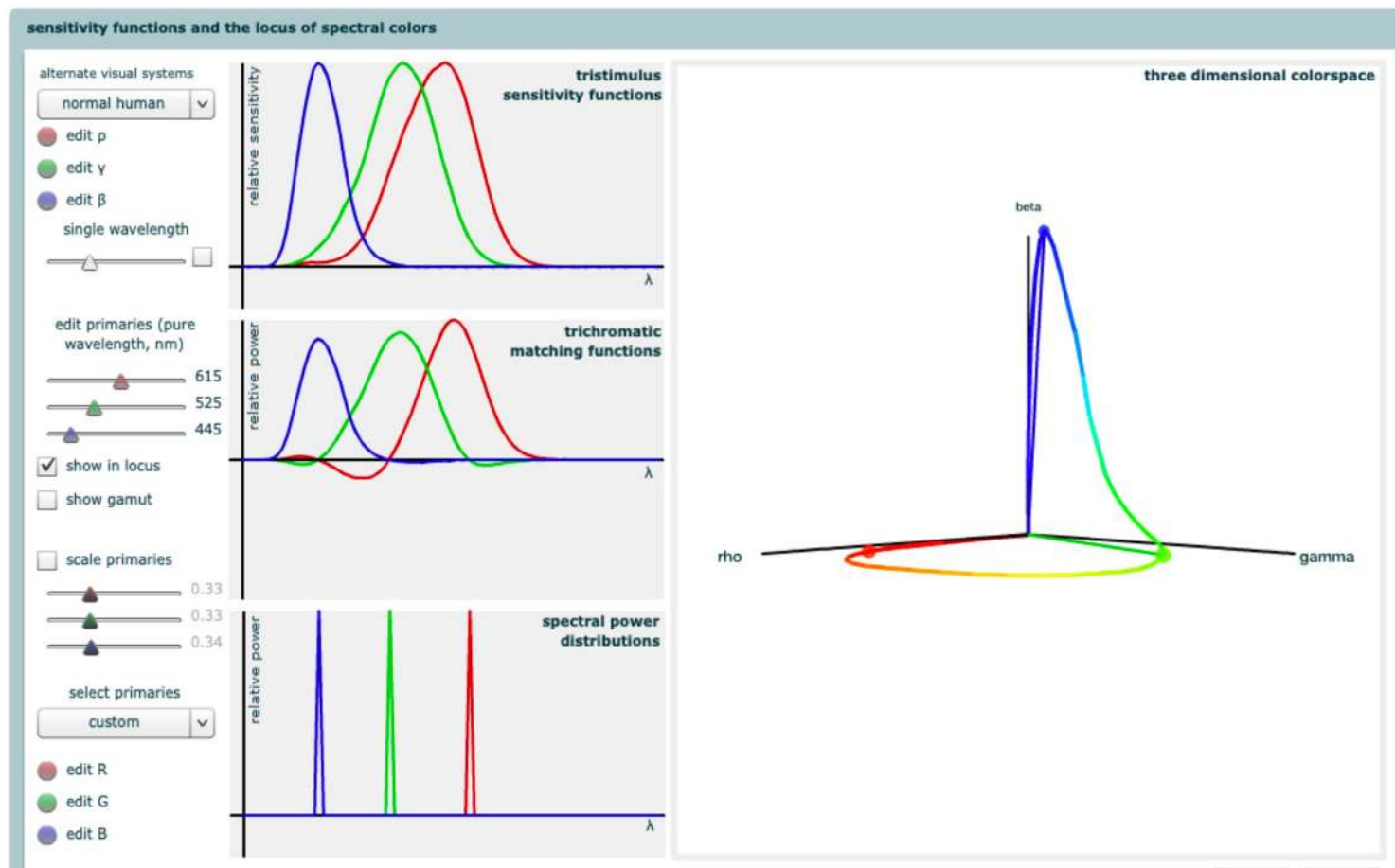
Locus of Spectral Colors / Spectral Locus



Locus.swf

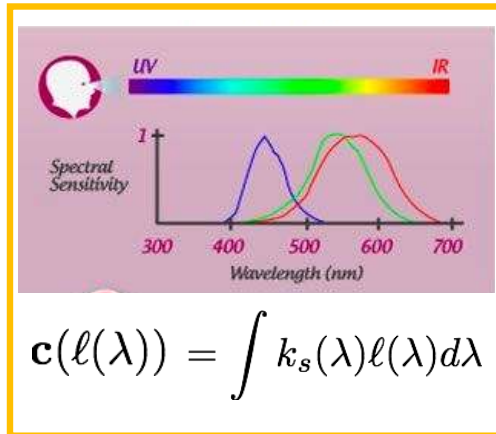


<https://sites.google.com/site/marclevoylectures/applets/color-theory>



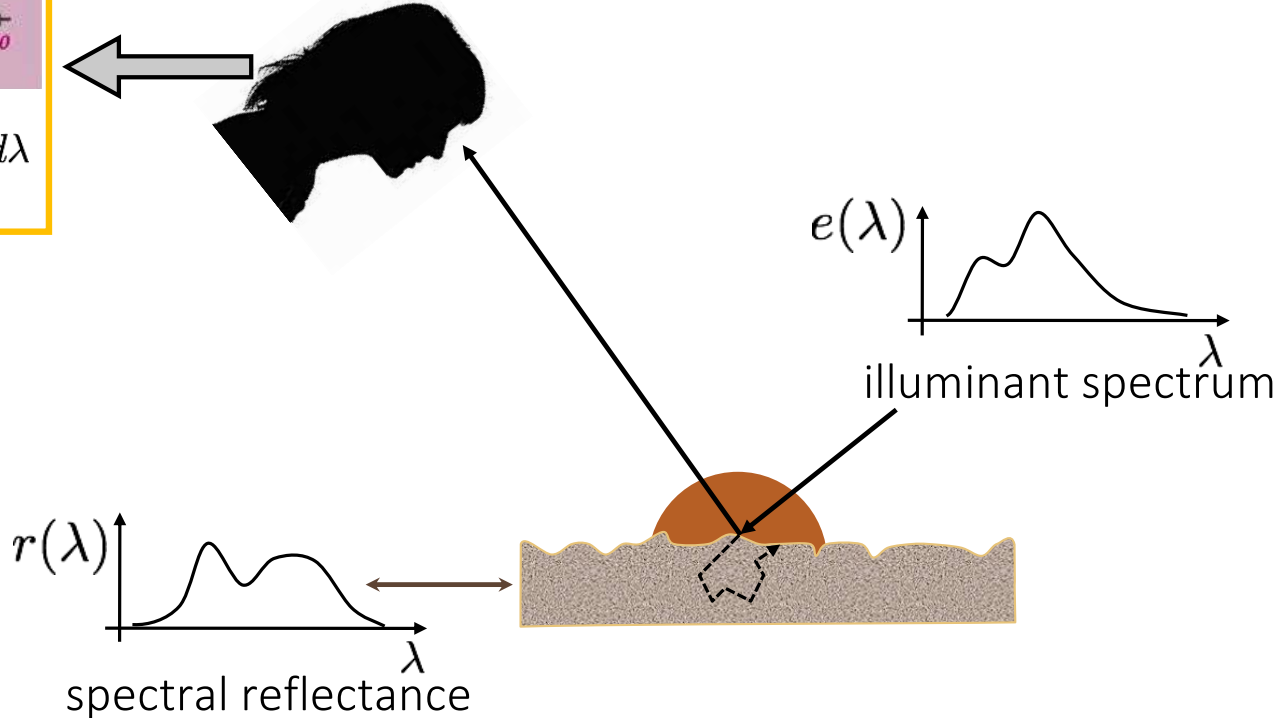
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Human color vision

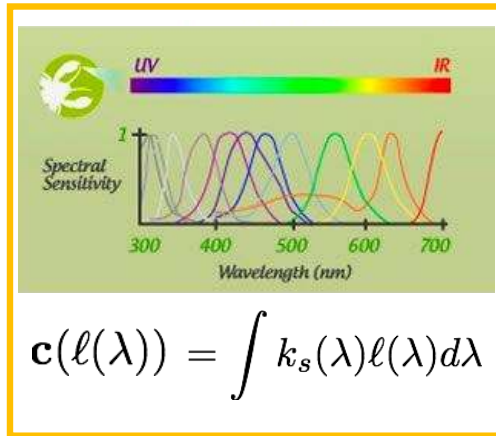


$$\ell(\lambda) = r(\lambda)e(\lambda)$$

spectral radiance

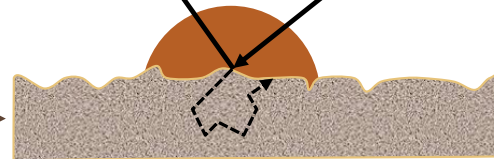
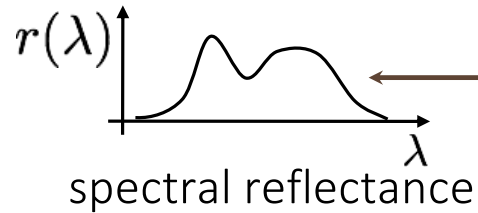
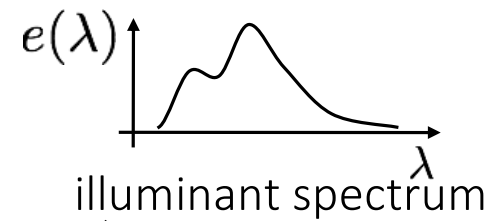
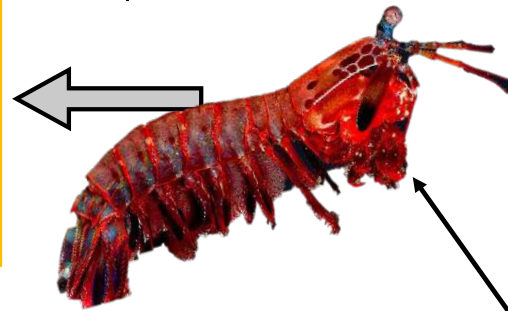


Mantis shrimp color vision



$$\ell(\lambda) = r(\lambda)e(\lambda)$$

spectral radiance



Human retinal color space

Does our brain see a different color for each different spectra in the visible light range of $\sim 400\text{-}700\text{ nm}$?



Yes



No

Spectral Sensitivity Function of Human Eye

$$S = \int_{\lambda} \Phi(\lambda) S(\lambda) d\lambda$$

$$M = \int_{\lambda} \Phi(\lambda) M(\lambda) d\lambda$$

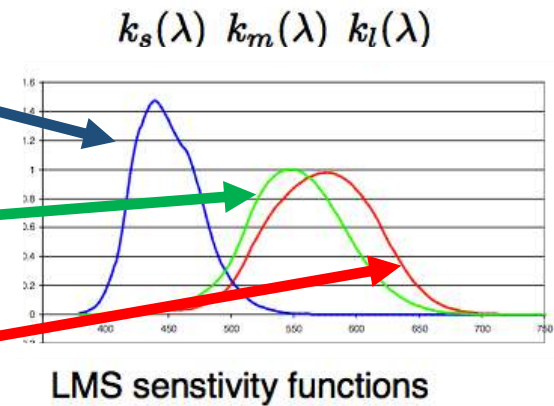
$$L = \int_{\lambda} \Phi(\lambda) L(\lambda) d\lambda$$

Spectral Sensitivity Function of Human Eye

$$S = \int_{\lambda} \text{Luminance} \left(\begin{array}{c} \text{Graph of } S(\lambda) \text{ from } 400 \text{ to } 700 \text{ nm} \end{array} \right) S(\lambda) d\lambda$$

$$M = \int_{\lambda} \text{Luminance} \left(\begin{array}{c} \text{Graph of } M(\lambda) \text{ from } 400 \text{ to } 700 \text{ nm} \end{array} \right) M(\lambda) d\lambda$$

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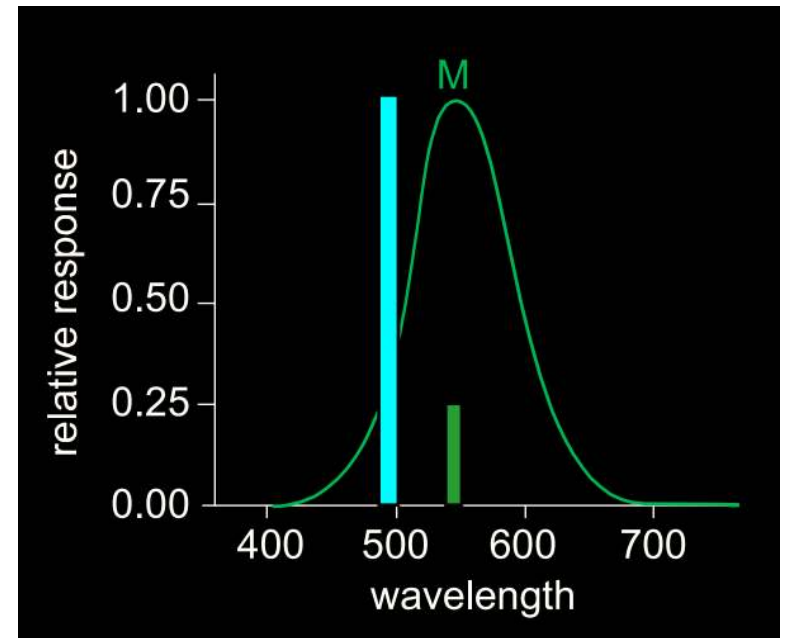


Spectral Sensitivity Function of Human Eye

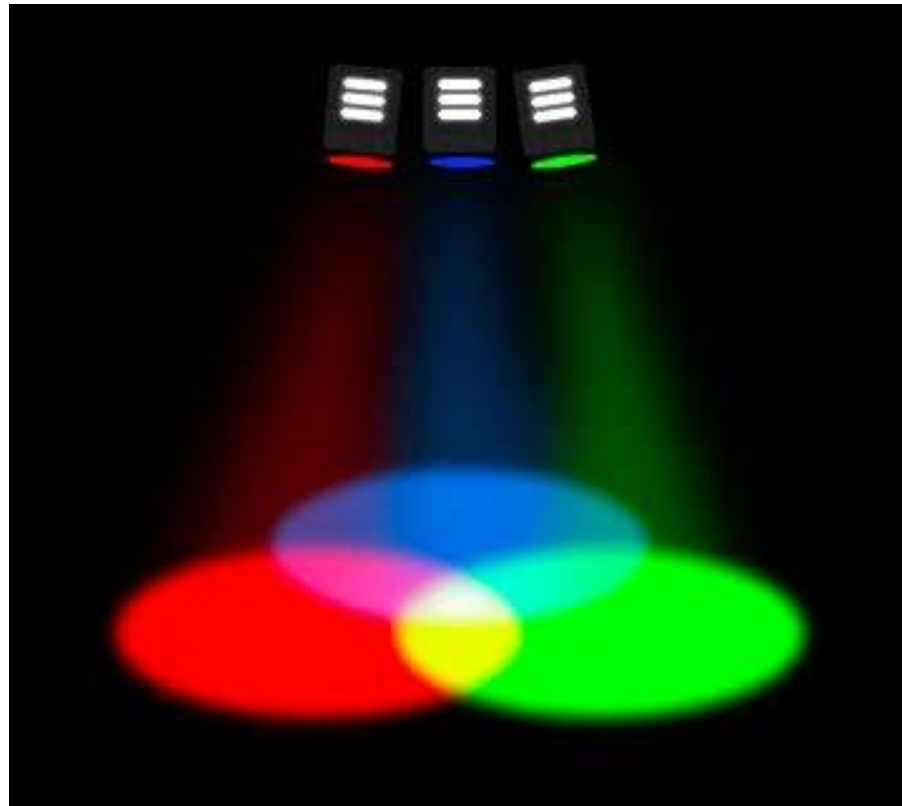
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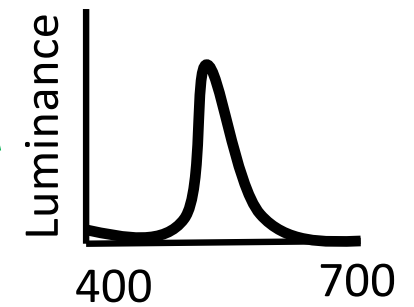
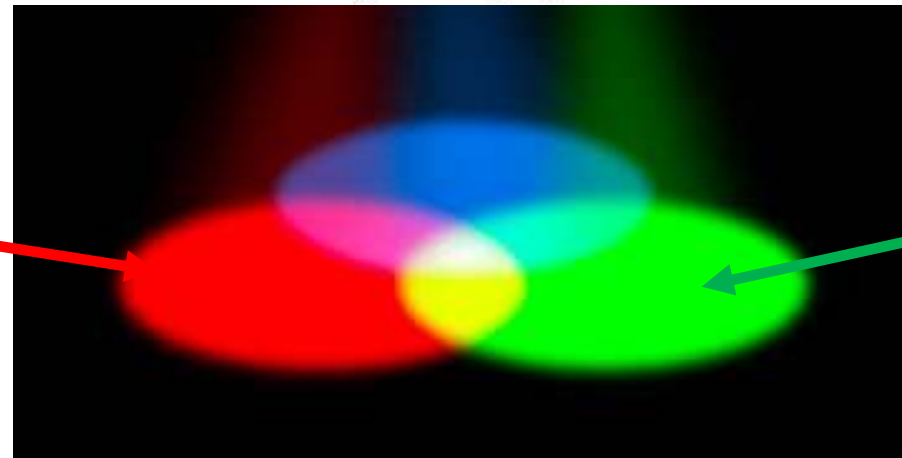
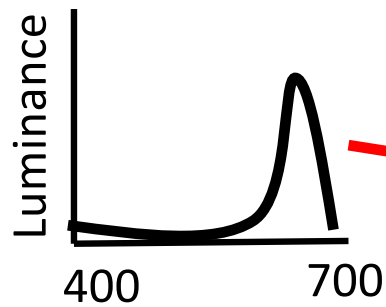
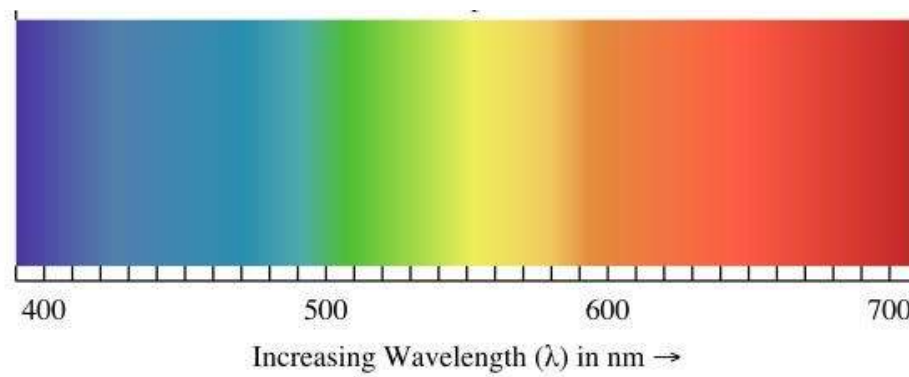
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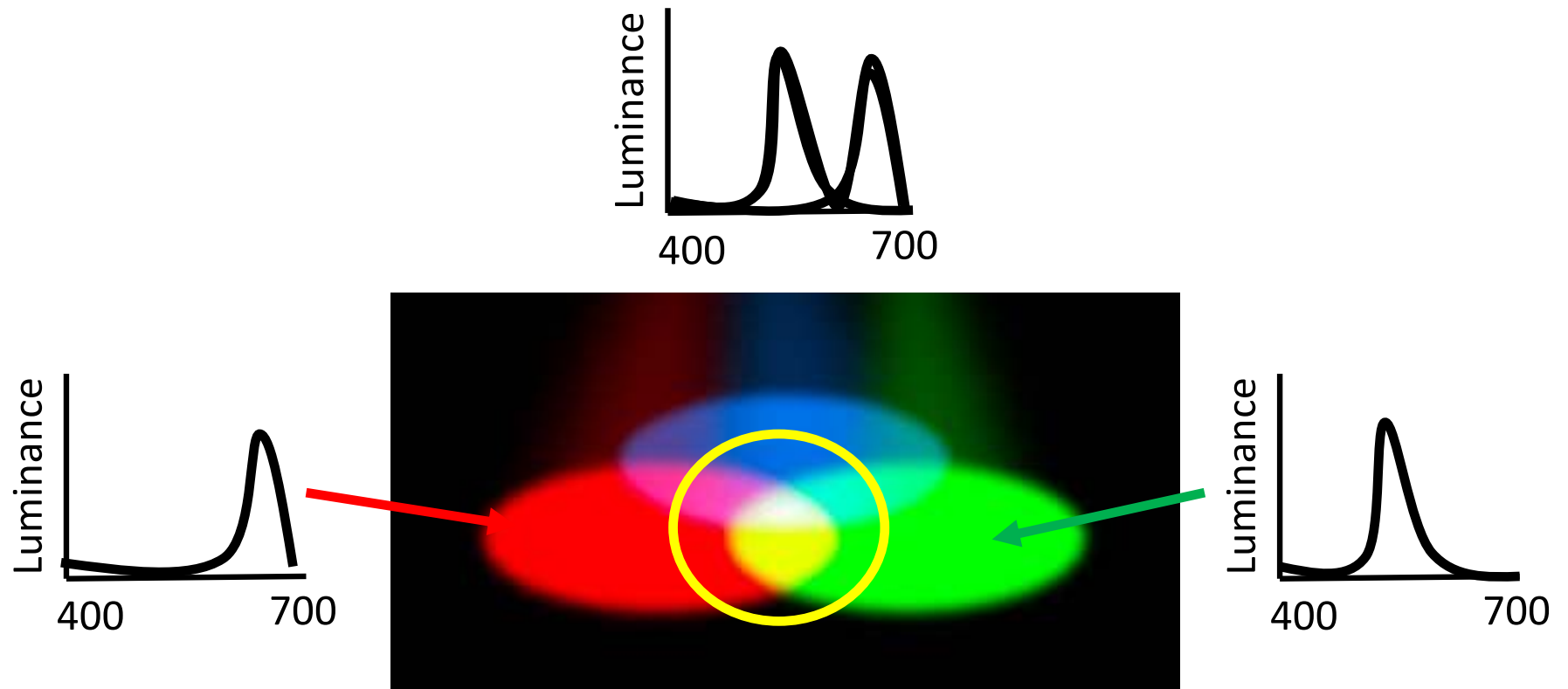
Light Mixing



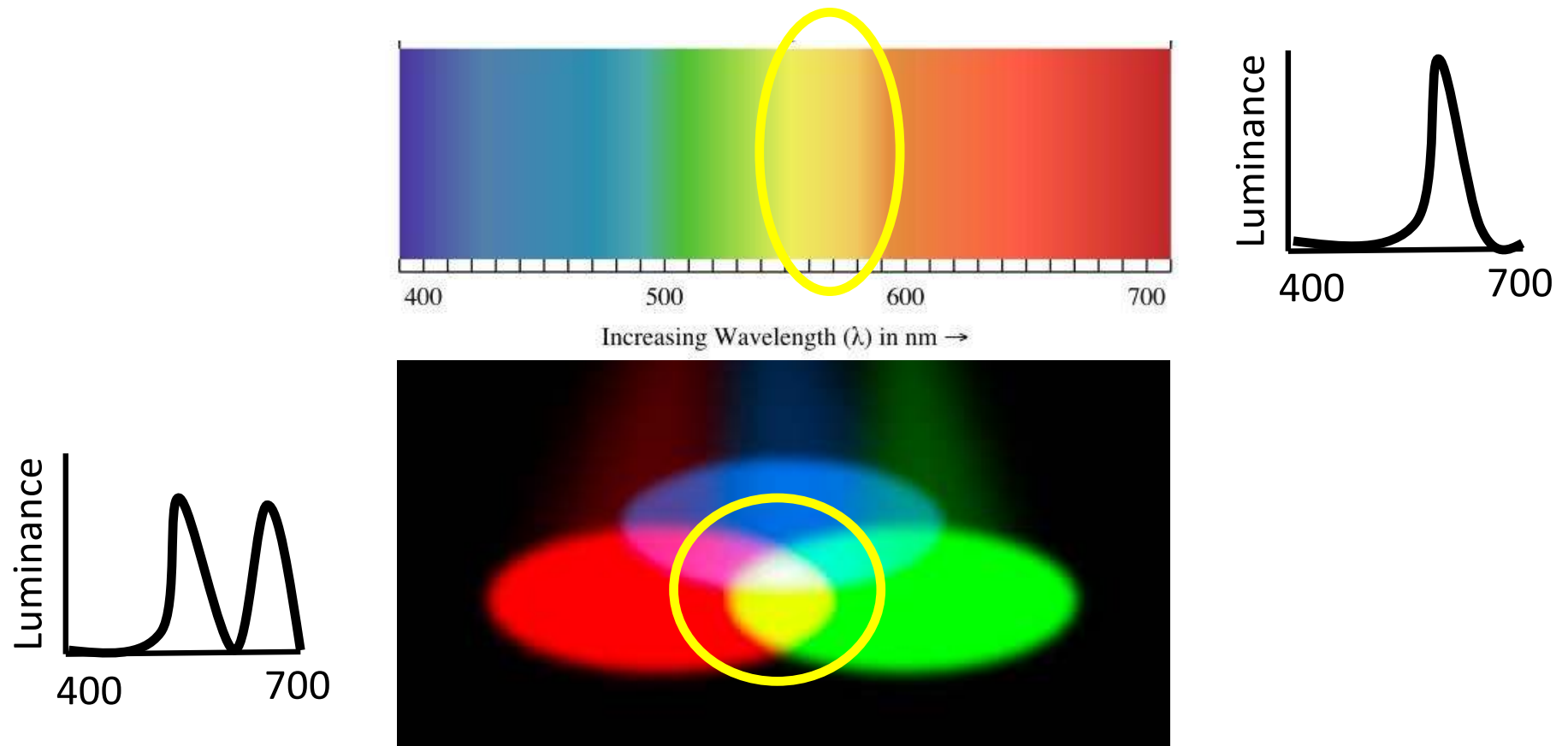
Light Mixing



Light Mixing



Spectral Sensitivity Function of Human Eye



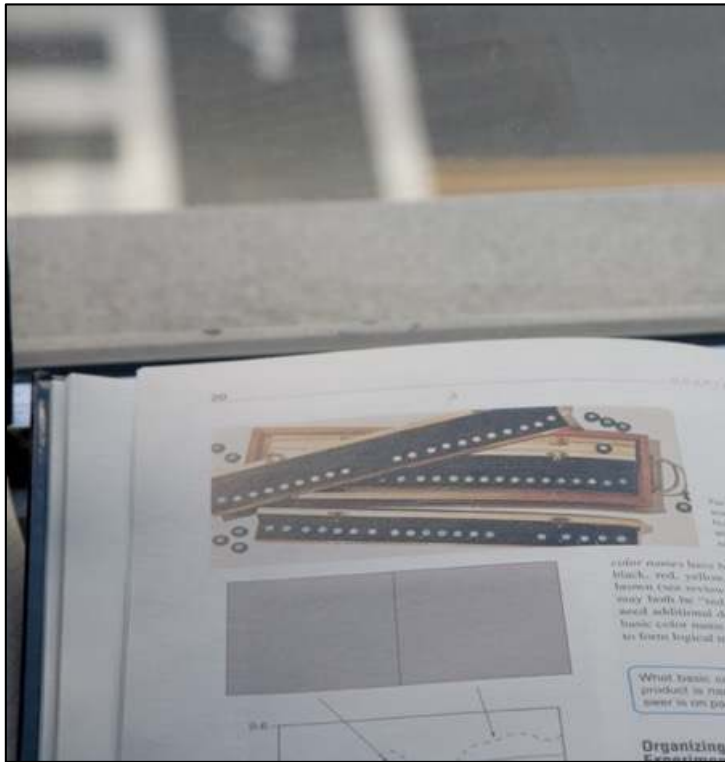
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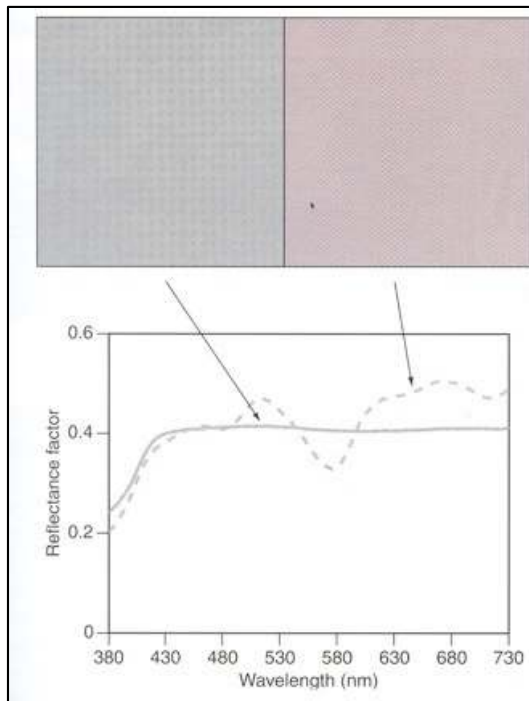
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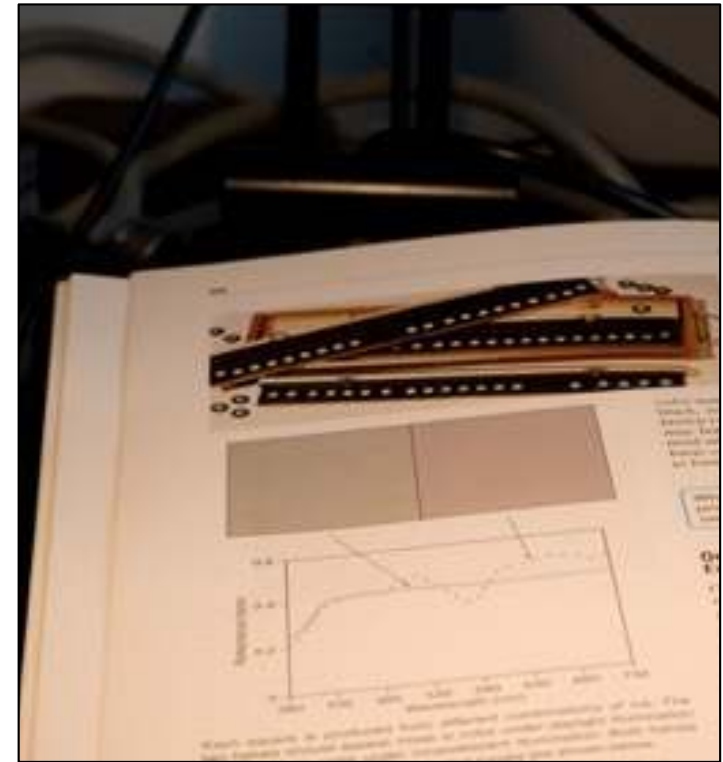
Illuminant metamerism



day light

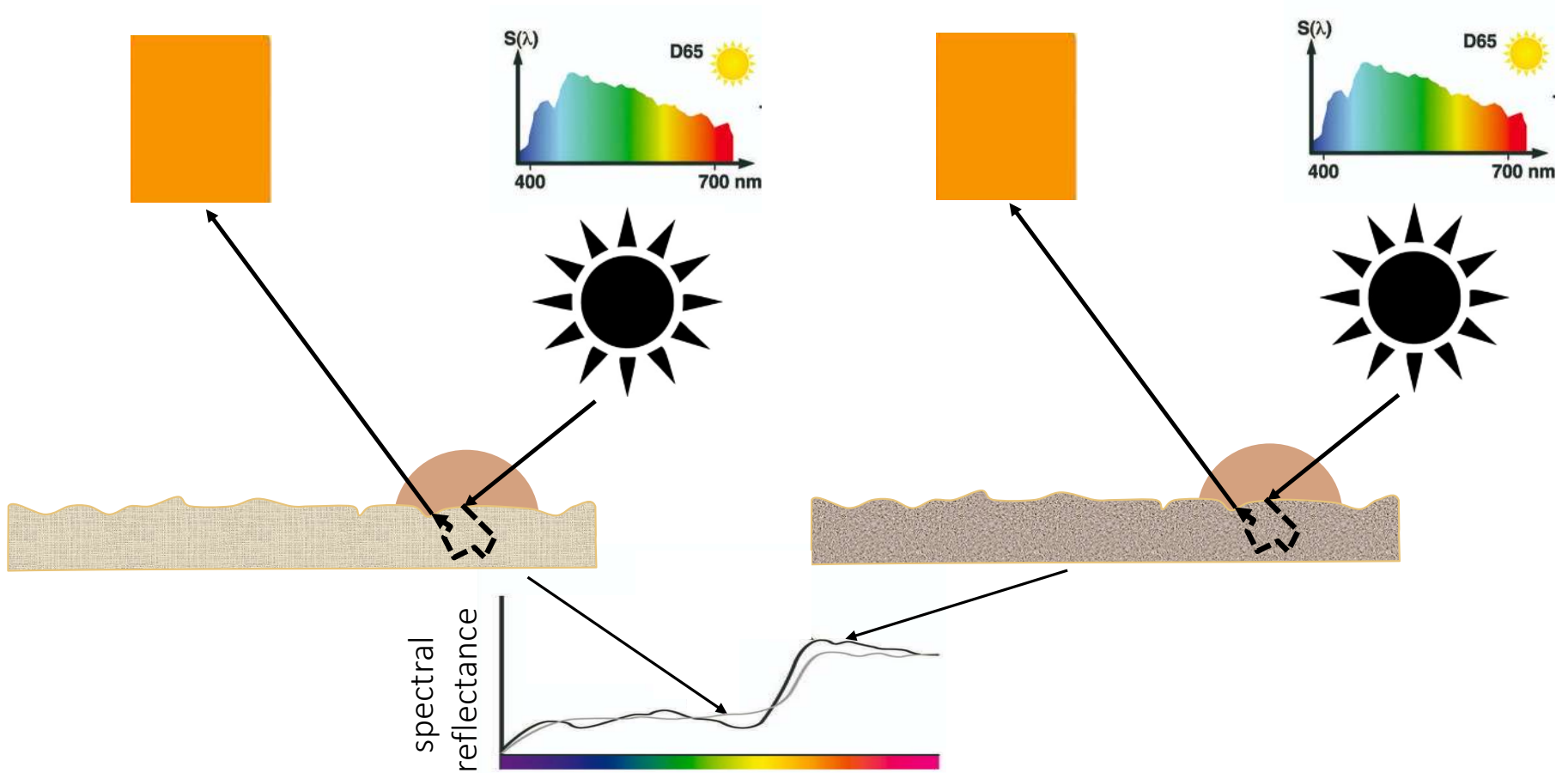


scanned copy

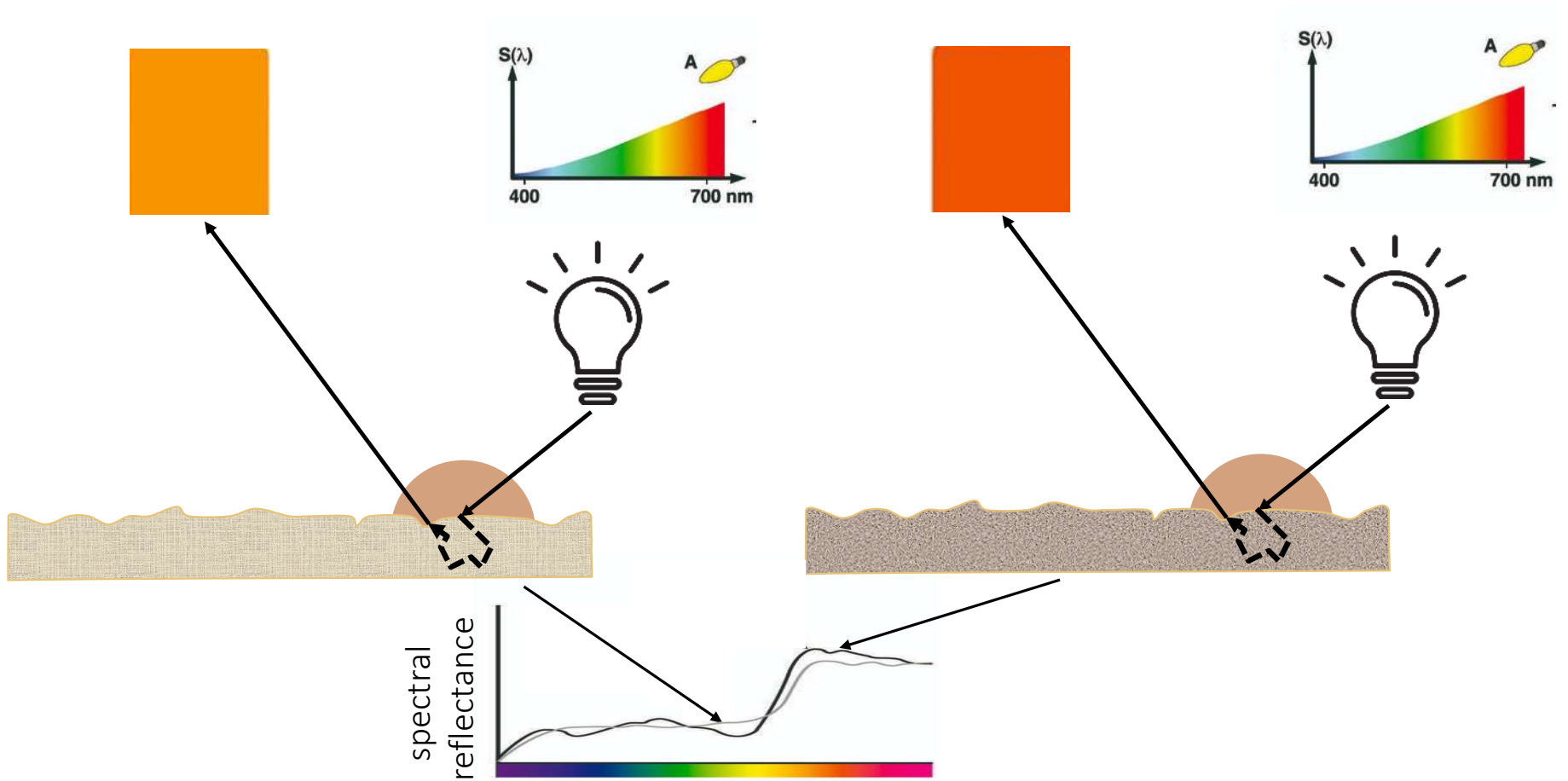


hallogen light

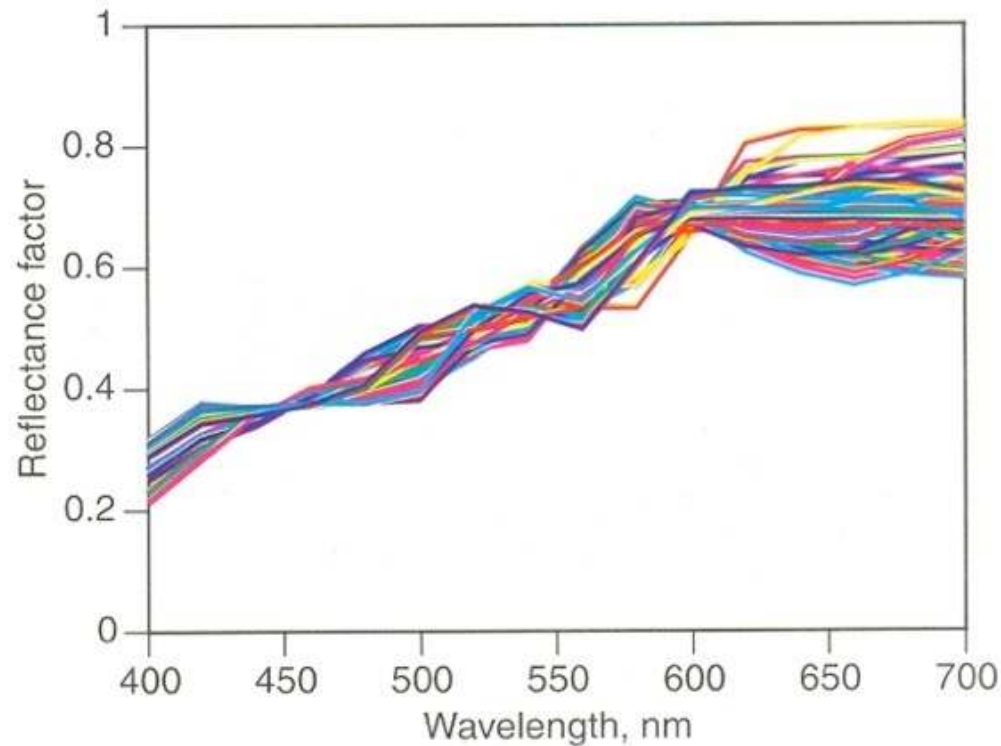
Illuminant metamerism



Illuminant metamerism



There is an infinity of metamers



Ensemble of spectral reflectance curves corresponding to three chromatic-pigment recipes all matching a tan material when viewed by an average observer under daylight illumination. [Based on Berns (1988b).]

The need for a well-agreed color space

- Around 1920s-1930s it became clear we needed a well-agreed color space
 - Motivation: they wanted to develop a colorimetric description of colored glass used for traffic control
- Why not measure the entire spectrum?
 - spectrophotometer hadn't been invented yet
 - different spectra can look exactly the same...so unnecessary specifications
- Why not measure cone sensitivity and use that?
 - Less directly measurable
 - The LMS cone responses were unknown when colorimetry was invented ~1931 (not measured until 1965)

Color matching

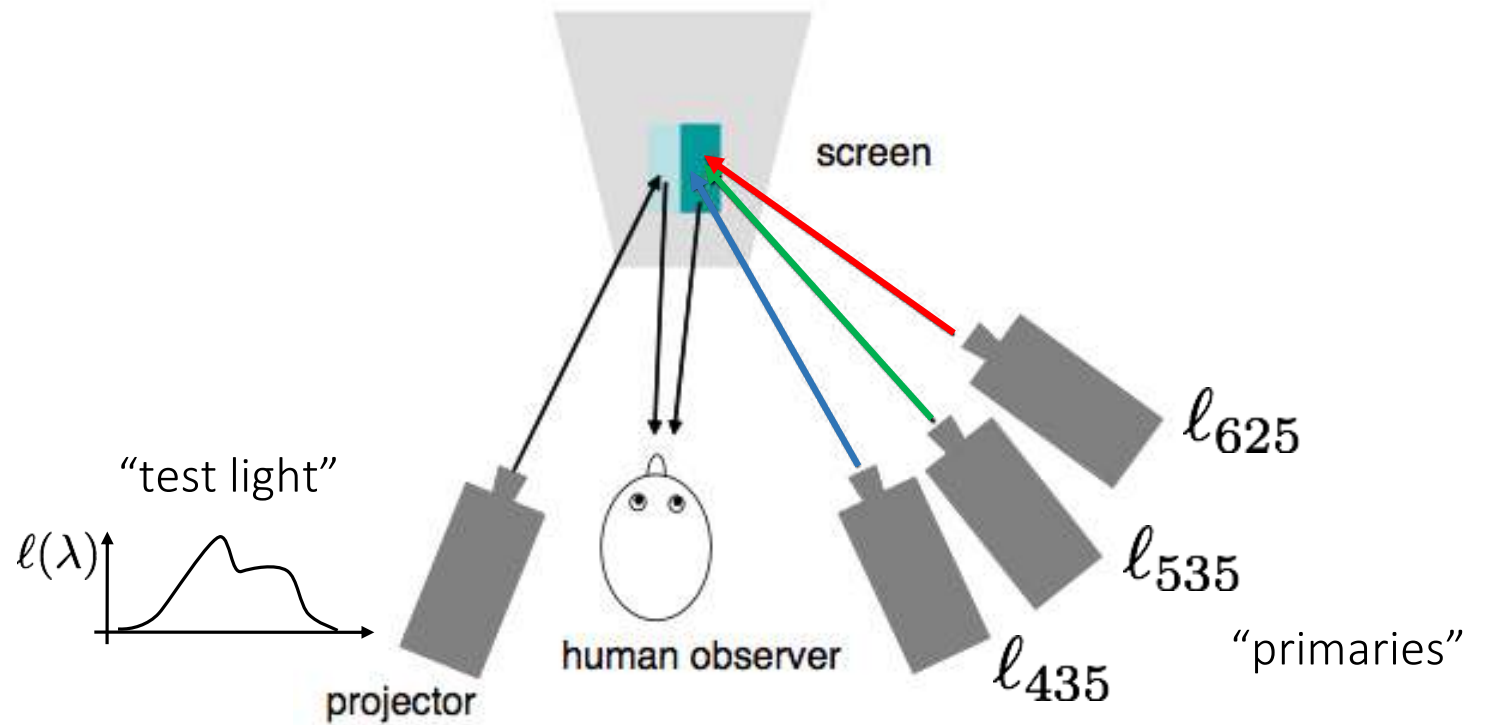
Color Matching



W. David Wright
Imperial College of London



John Guild
National Physical Laboratory

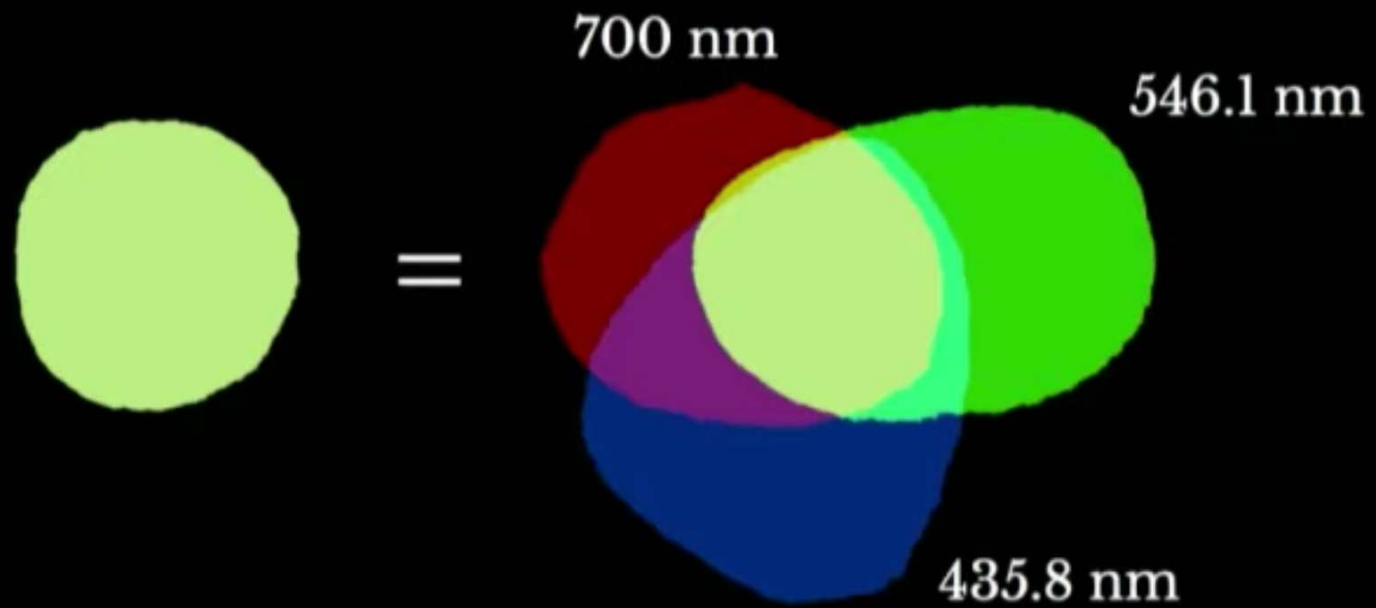


Color Matching



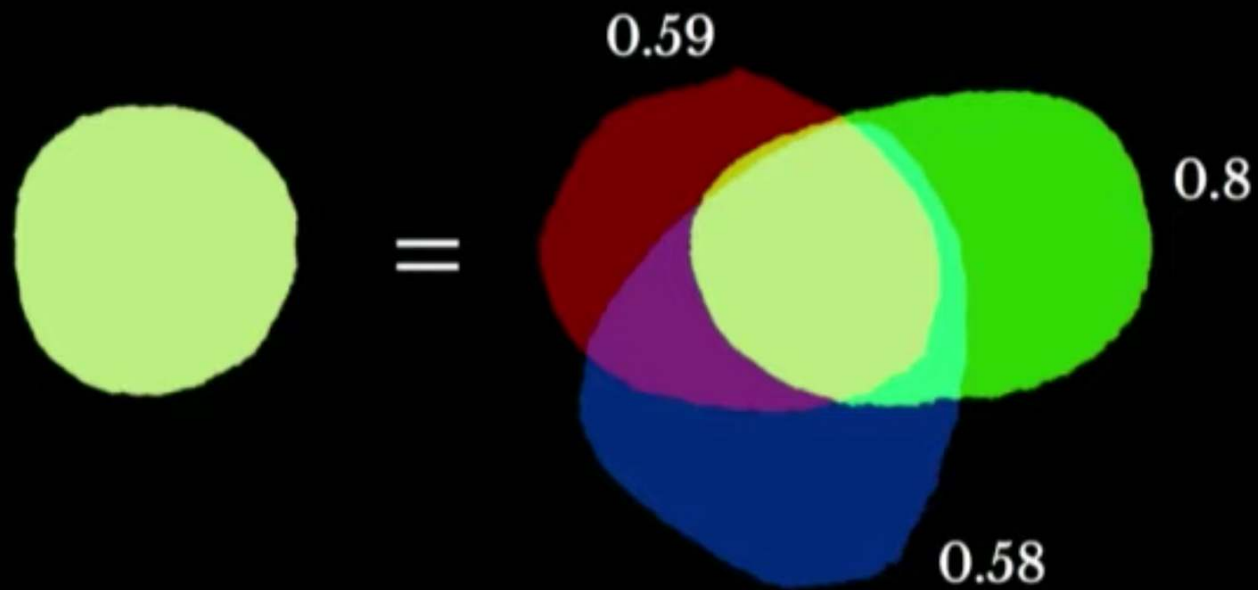
"RGB to XYZ: The Science and History of Color" by John Austin

Color Matching



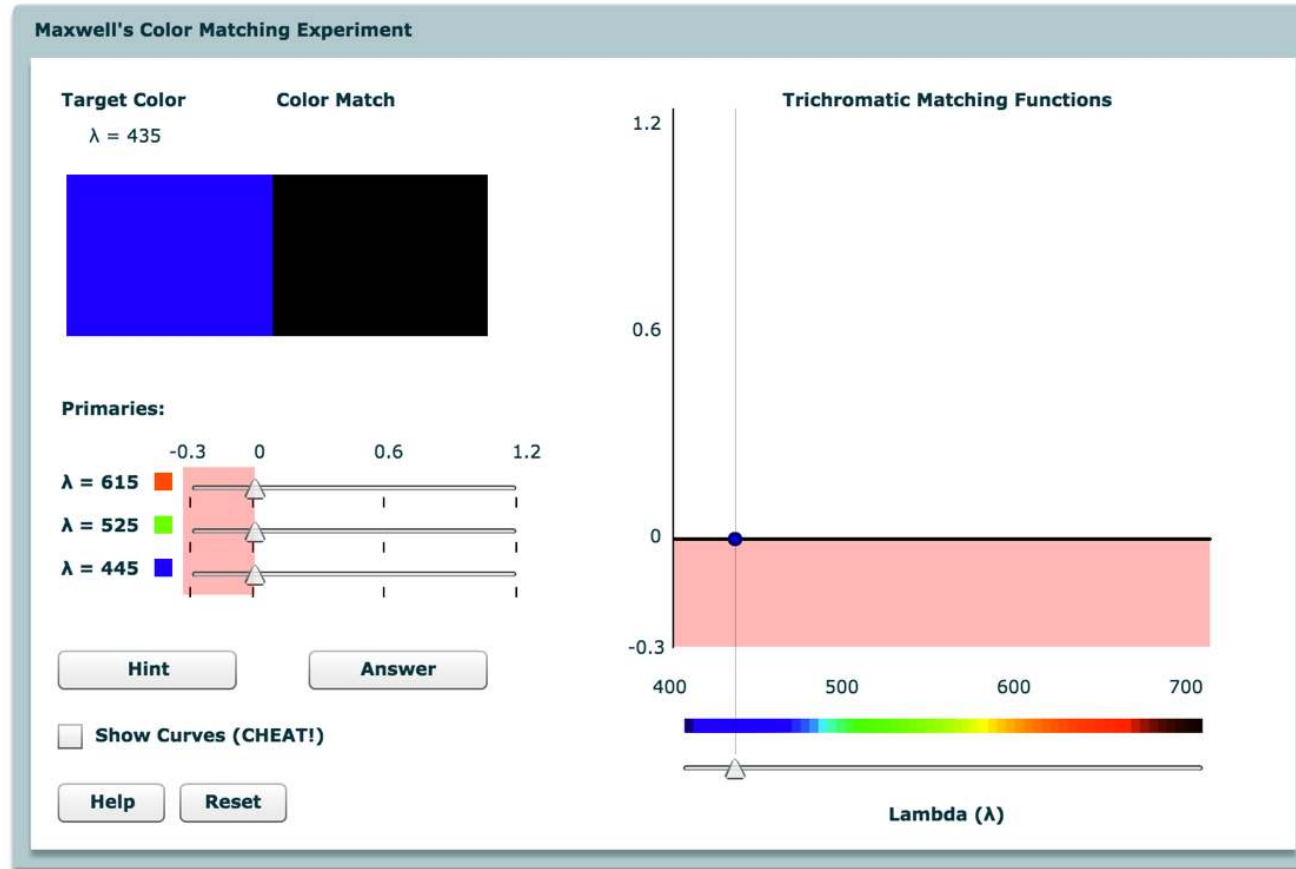
"RGB to XYZ: The Science and History of Color" by John Austin

Color Matching



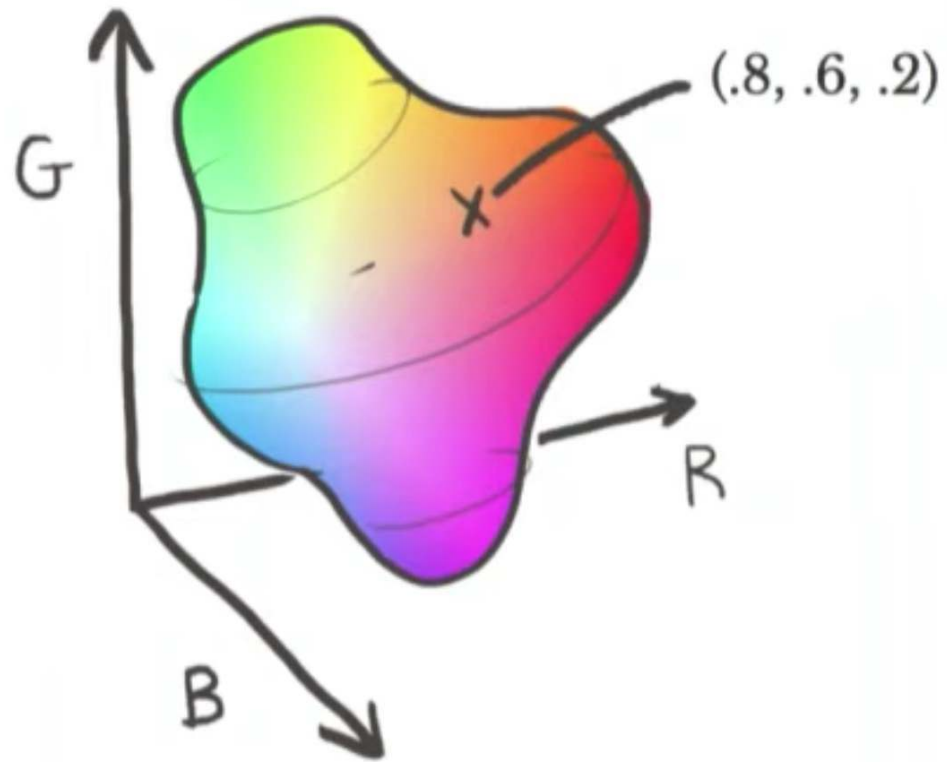
"RGB to XYZ: The Science and History of Color" by John Austin

Color matching demo



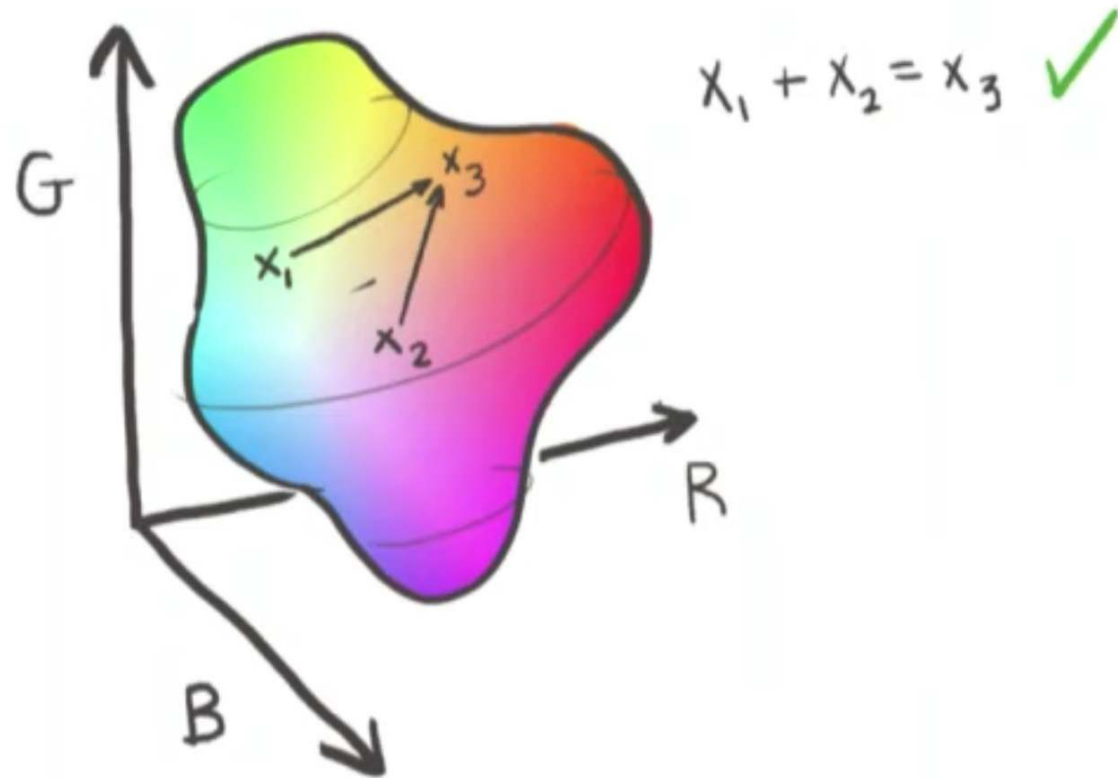
<http://graphics.stanford.edu/courses/cs178/applets/colormatching.html>

3D Color Space



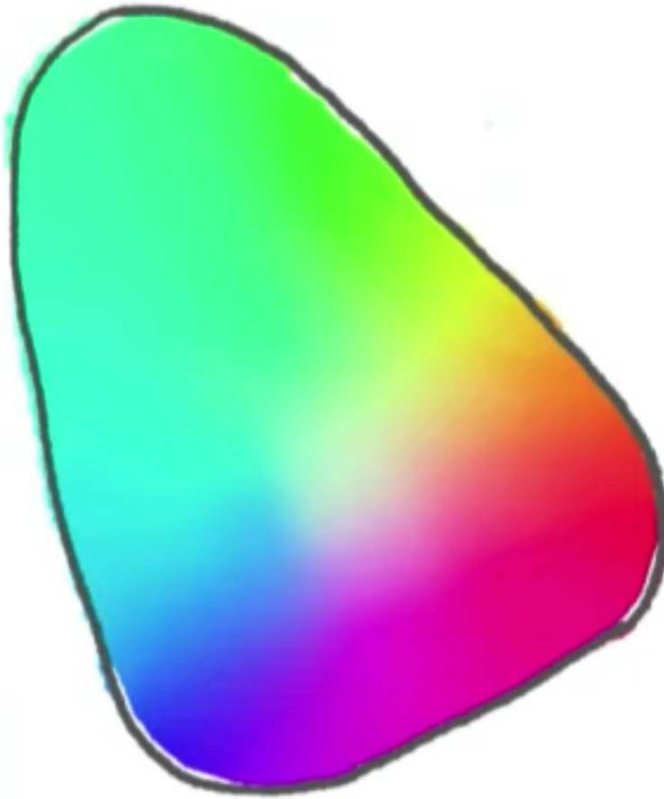
"RGB to XYZ: The Science and History of Color" by John Austin

3D Color Space : Additive Colors



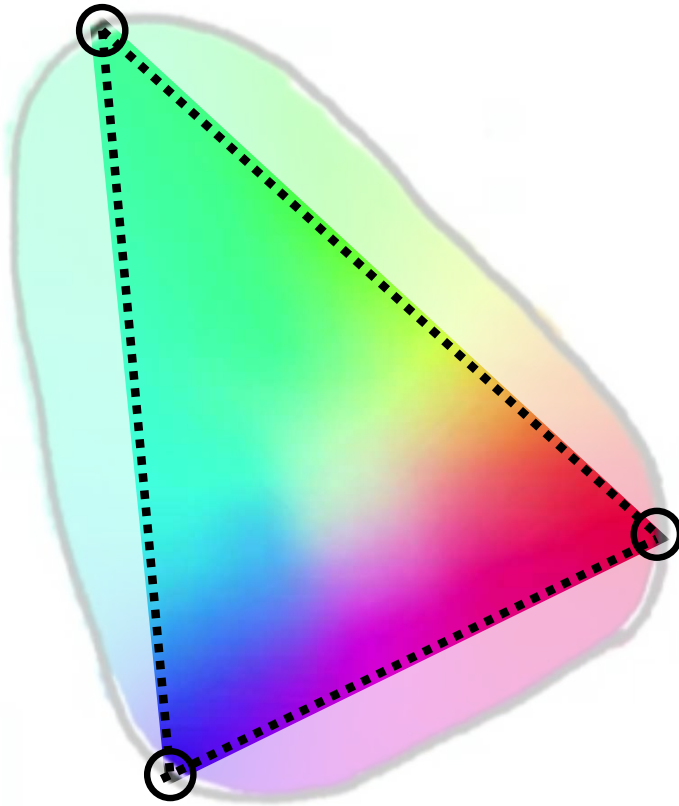
Colors are Linear

Projected 2D Color Space



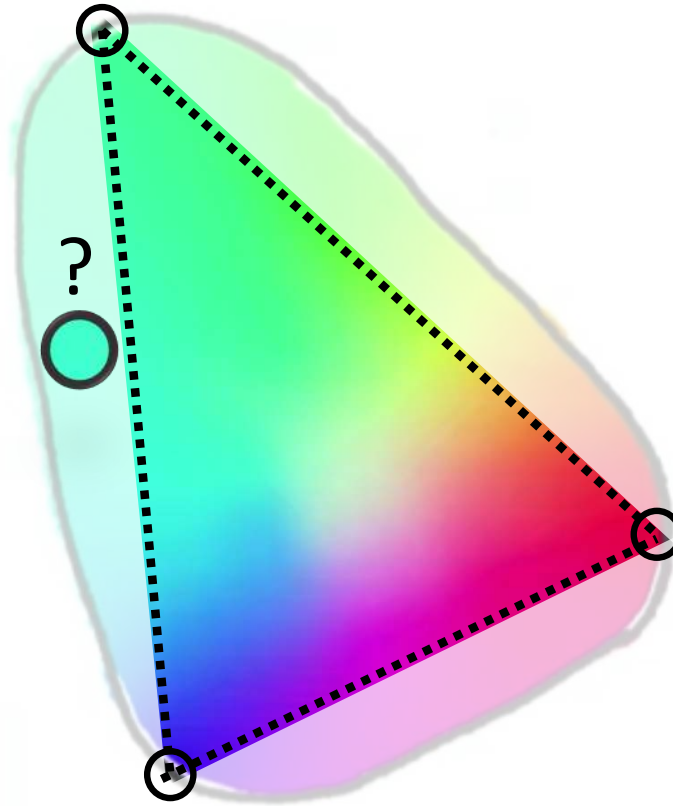
(note: not an accurate representation)

Projected 2D Color Space



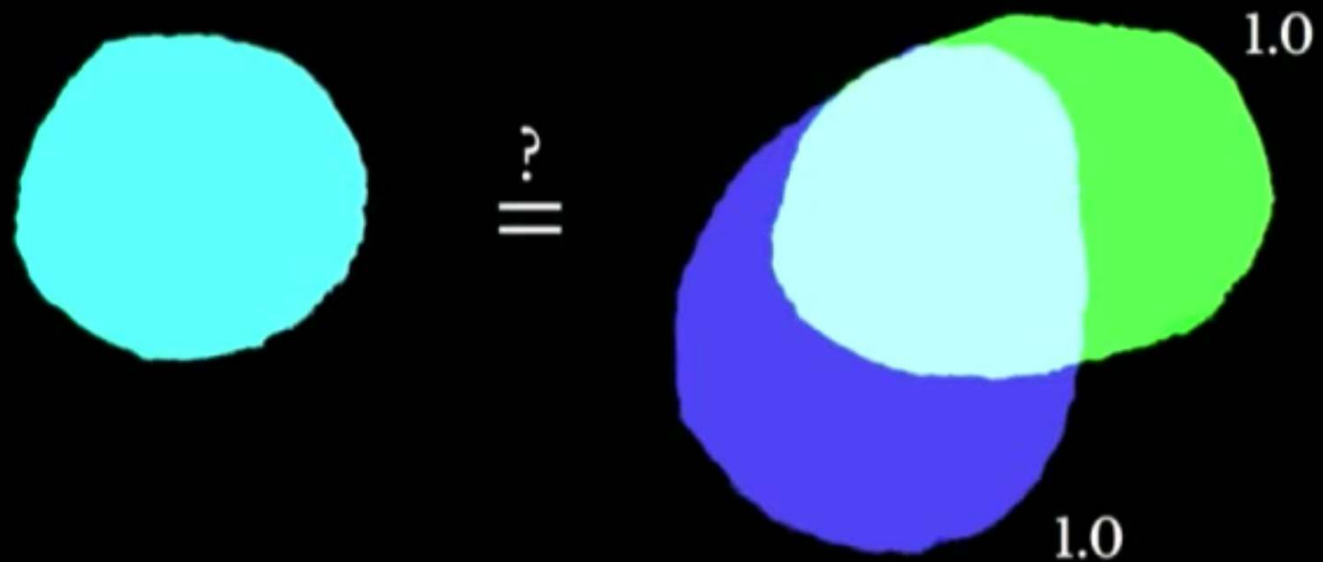
(note: not an accurate representation)

Projected 2D Color Space



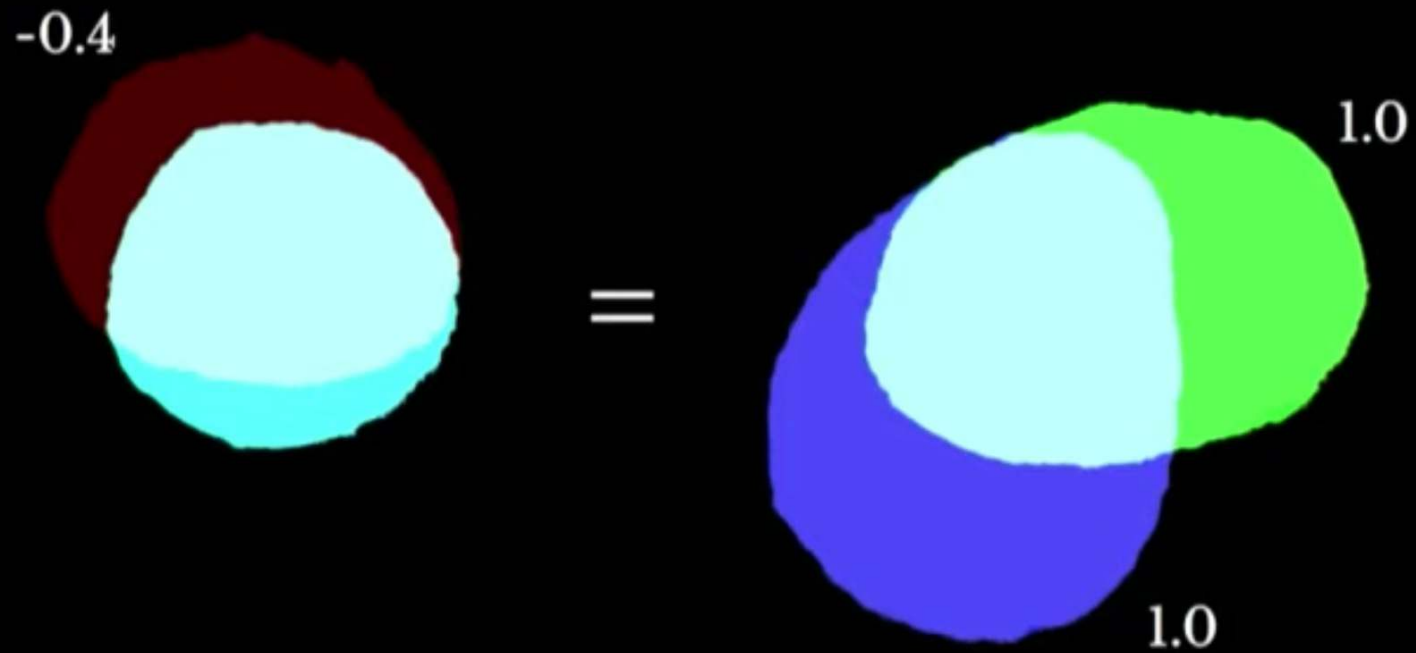
(note: not an accurate representation)

Color Matching



"RGB to XYZ: The Science and History of Color" by John Austin

Color Matching



"RGB to XYZ: The Science and History of Color" by John Austin

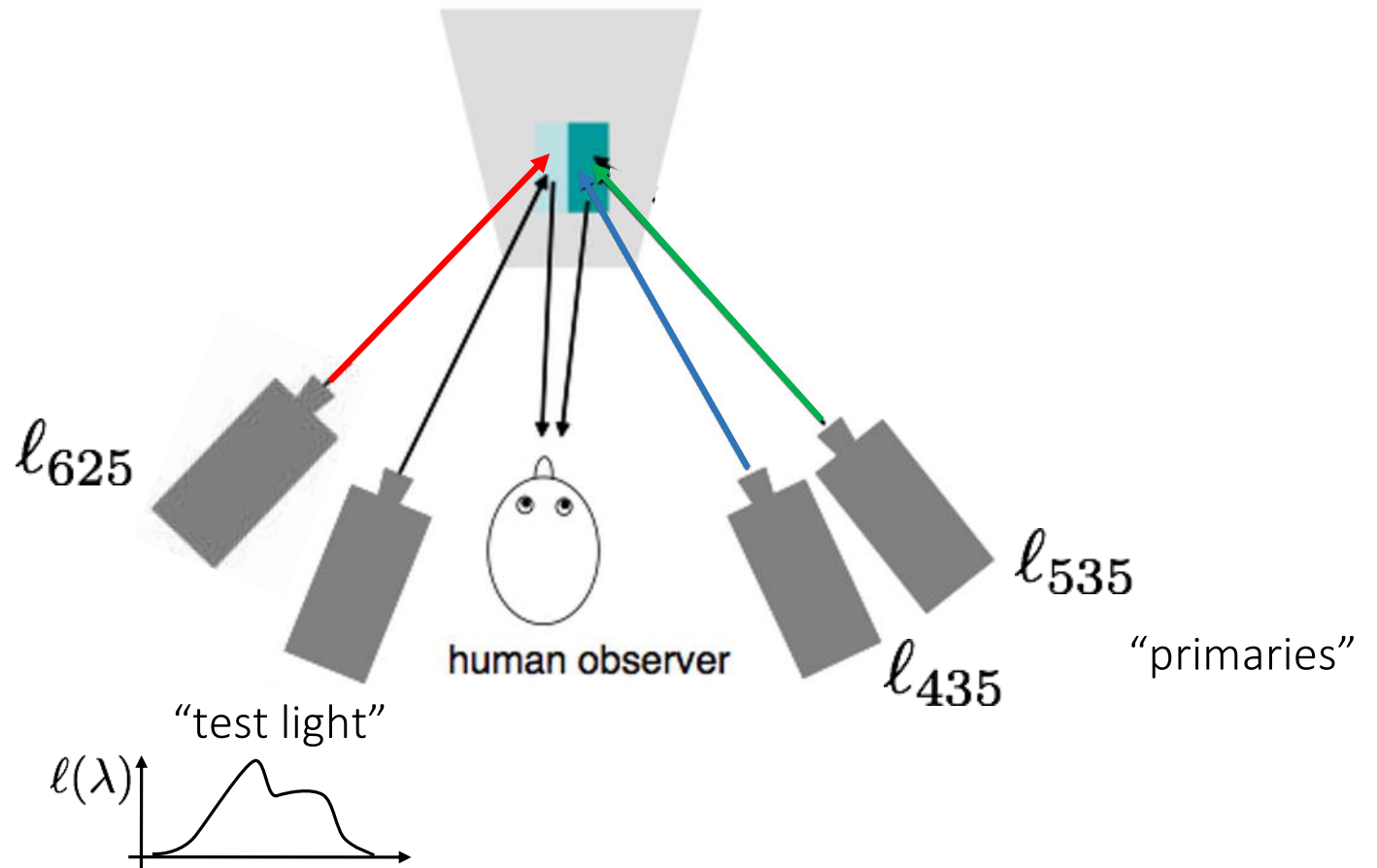
Color Matching



W. David Wright
Imperial College of London



John Guild
National Physical Laboratory



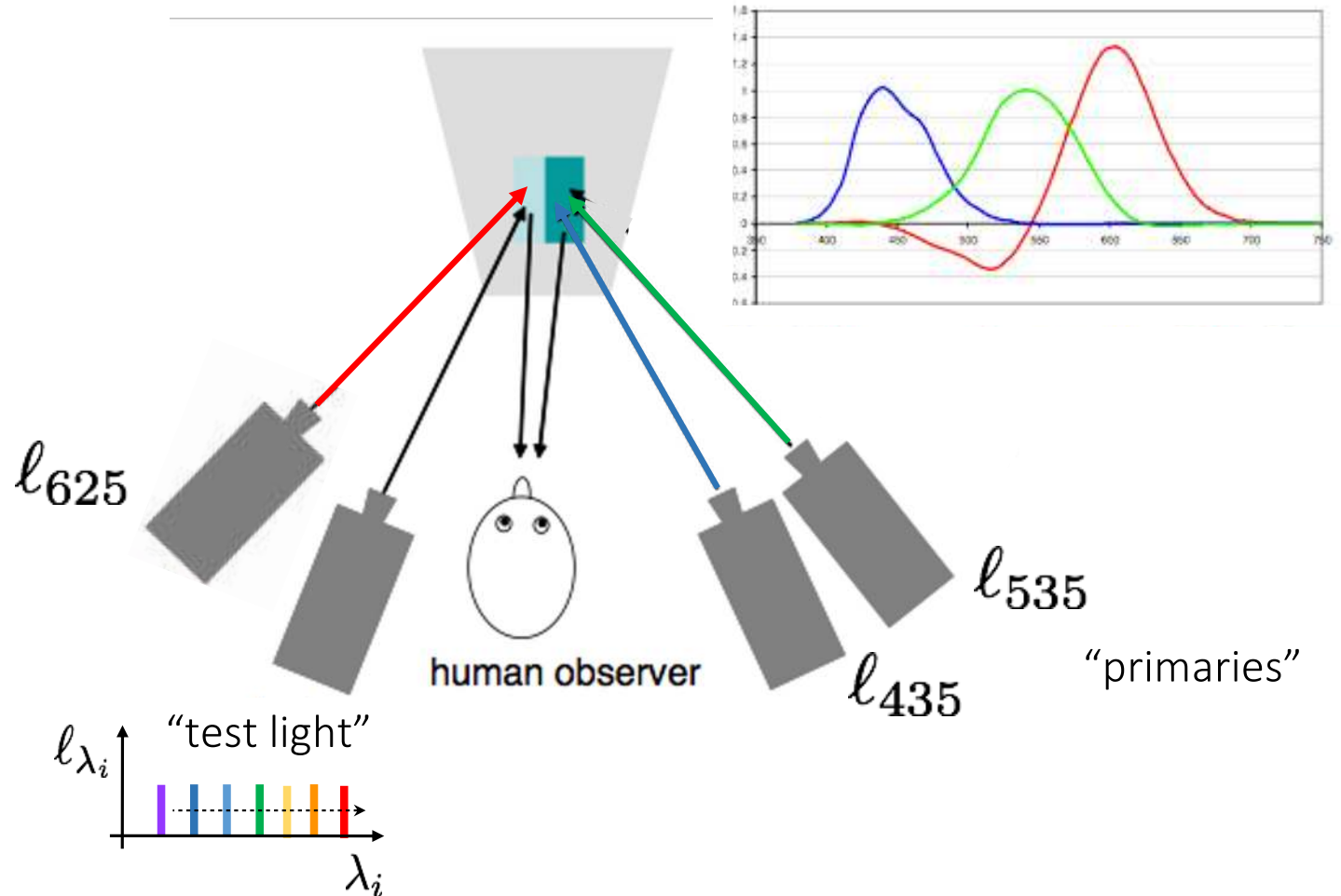
Color Matching



W. David Wright
Imperial College of London



John Guild
National Physical Laboratory



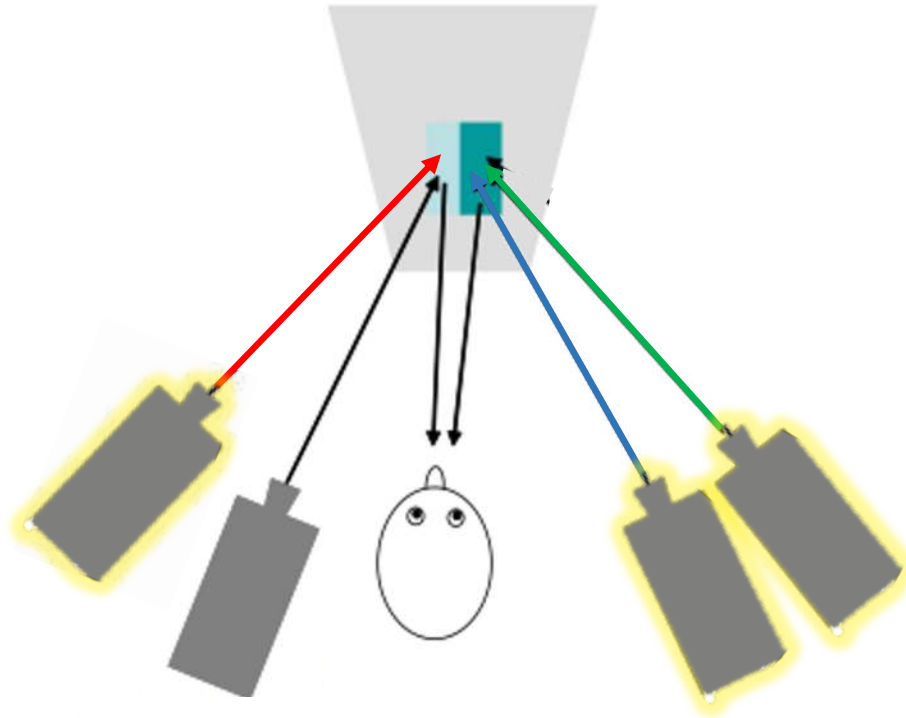
Color Matching



W. David Wright
Imperial College of London
(10 observers)



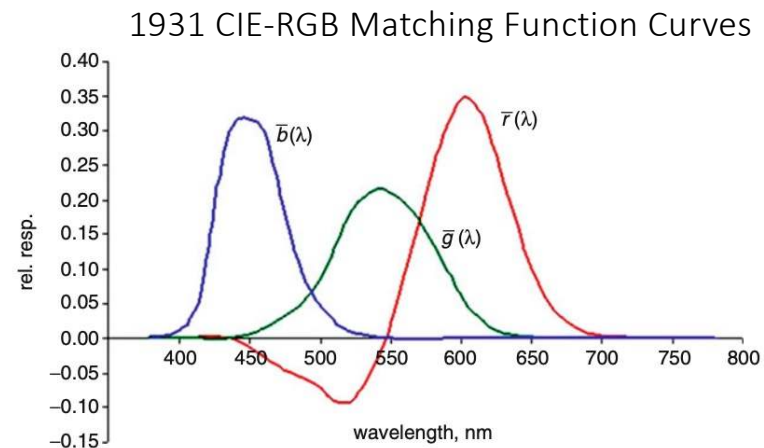
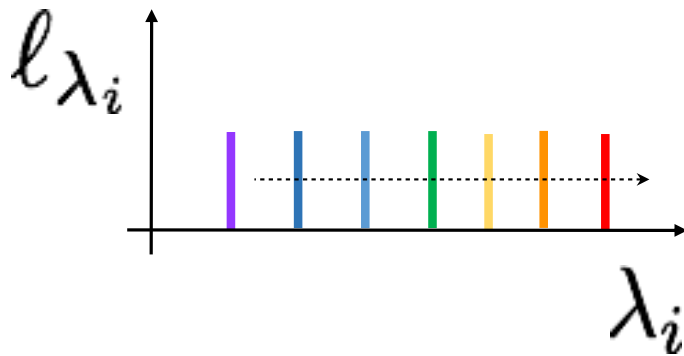
John Guild
National Physical Laboratory
(7 observers)



Despite using different primaries they matched when put
in a common space!

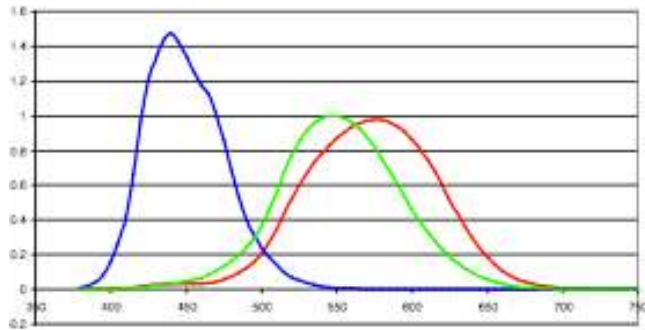
CIE Color Matching

- An organization called Commission Internationale de l'Eclairage (CIE) defined all practical standards for color measurements (colorimetry)
- CIE 1931 Standard – uses color patches that subtend 2 degrees of visual angle and uses color primaries defined as pure line spectra at 700 nm, 546.1 nm, 435.8 nm



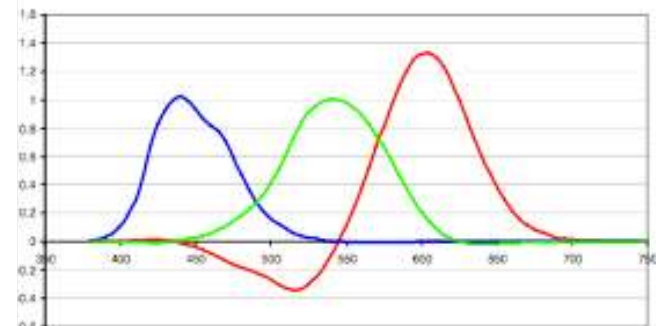
Two views of retinal color

Human Retinal Color Curves



Retinal color is produced by analyzing spectral power distributions using the color sensitivity functions.

CIE-RGB Matching Function Curves

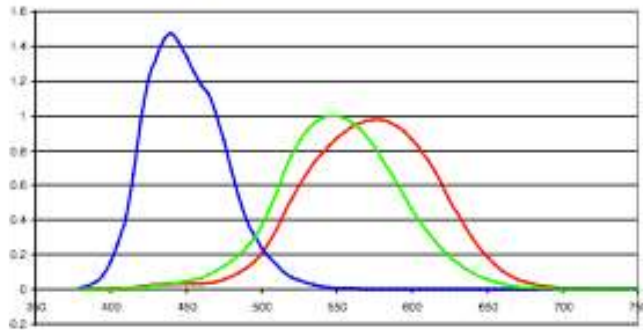


Retinal color is produced by synthesizing color primaries using the color matching functions.

What is the advantage of the CIE-RGB color curves?

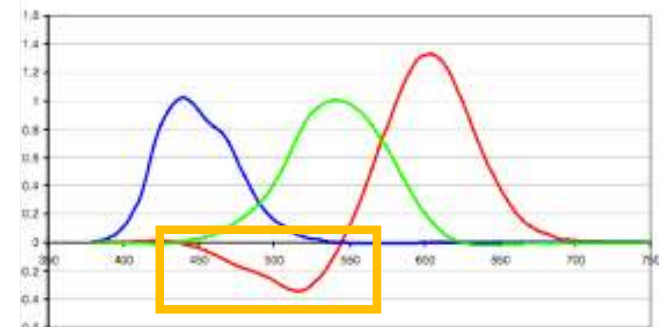
Two views of retinal color

Human Retinal Color Curves



Retinal color is produced by analyzing spectral power distributions using the color sensitivity functions.

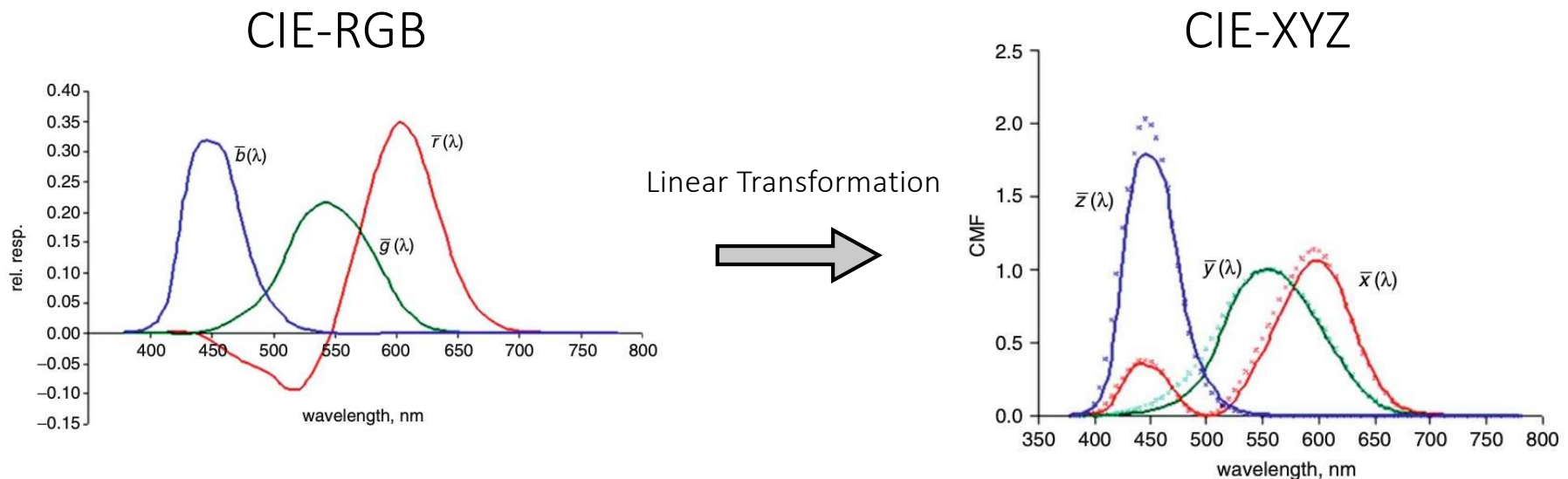
CIE-RGB Matching Function Curves



Retinal color is produced by synthesizing color primaries using the color matching functions.

What is the advantage of the human retinal color curves?

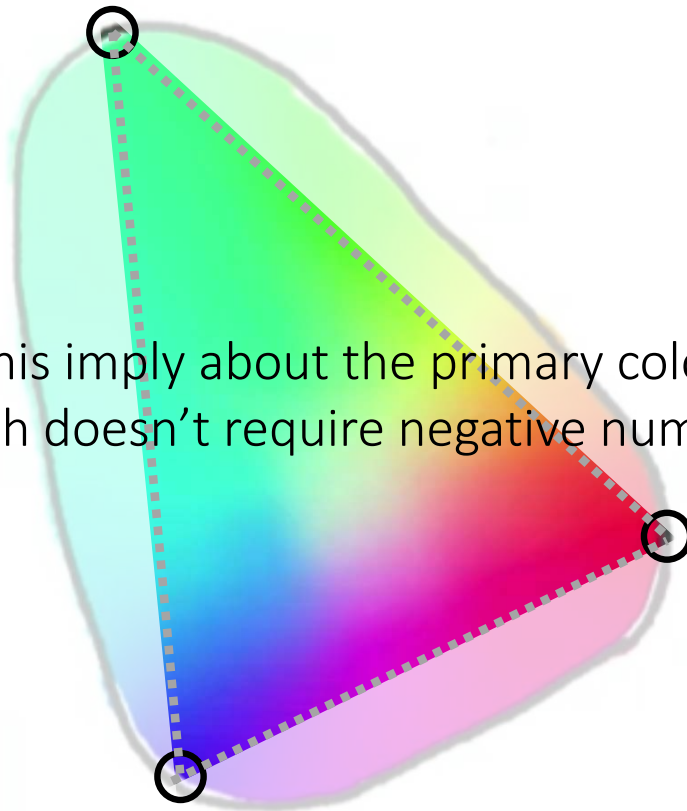
CIE-RGB $\rightarrow$ CIE-XYZ



- Derived from CIE RGB by adding enough blue and green to make the red positive.
- Probably the most important *reference* (i.e., device independent) color space.

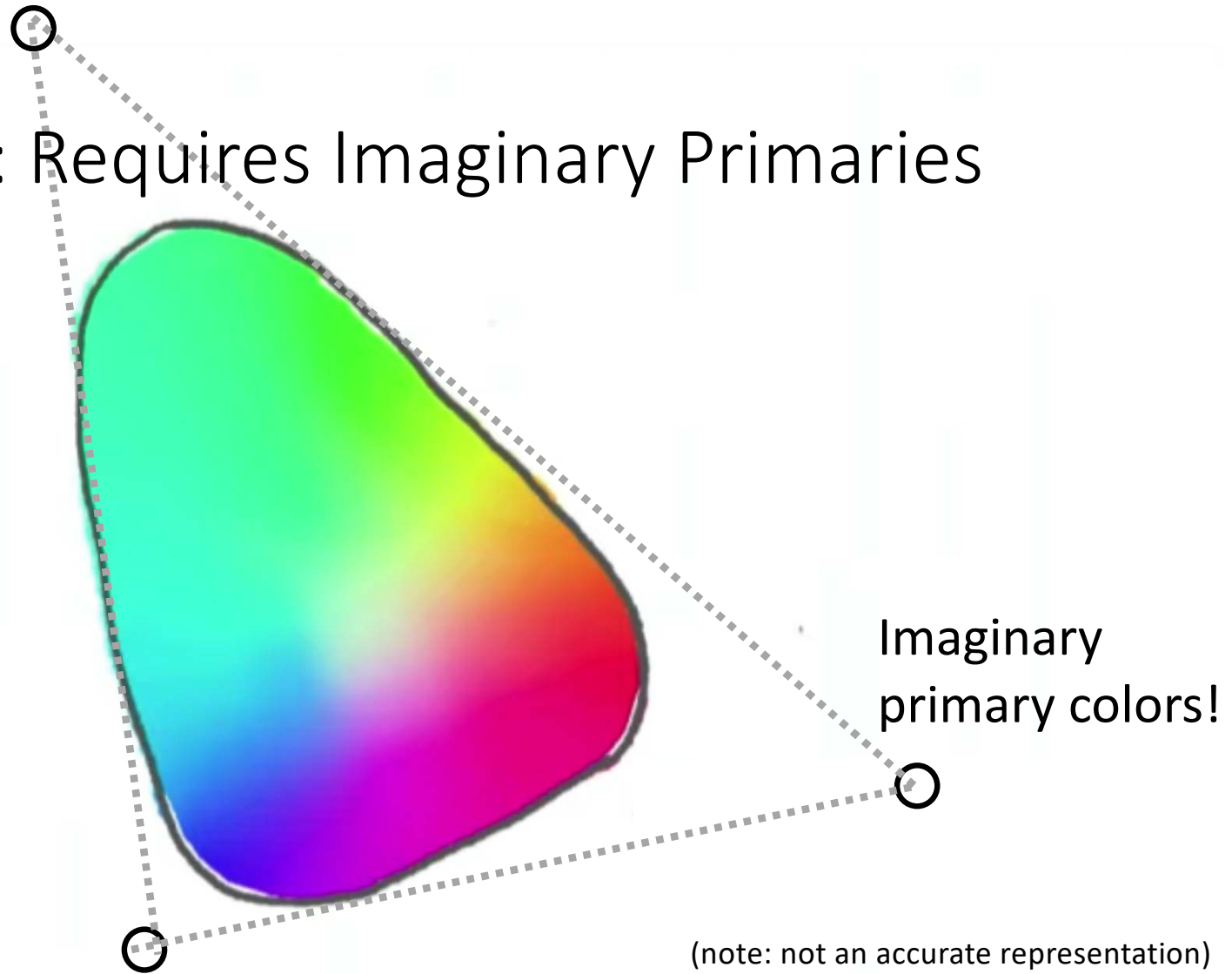
CIE-RGB : Requires Negative Light Primaries

What does this imply about the primary colors for CIE-XYZ
(which doesn't require negative numbers)?



(note: not an accurate representation)

CIE-XYZ : Requires Imaginary Primaries



Other Important Color Spaces

sRGB

(s stands for standard...
which ends up being not so standard)



*A (sometimes nonlinear) standardized
color space built for the digital world*

CIE-Lab

(L stands for luminance;
a controls green/red, b controls blue/yellow)

Other Important Color Spaces

sRGB

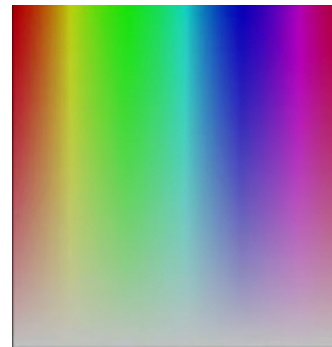
(s stands for standard...
which ends up being not so standard)



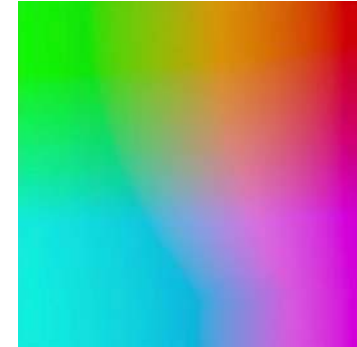
A (sometimes nonlinear) standardized color space built for the digital world

CIE-Lab

(L stands for luminance;
a controls green/red, b controls blue/yellow)



Constant brightness slice
in linear space



Constant L slice in
nonlinear CIE-Lab

*Nonlinear function of CIE-XYZ (RGB),
but is a perceptually uniform space*

"RGB to XYZ: The Science and History of Color" by John Austin

Other Important Color Spaces

sRGB

(s stands for standard...
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A (sometimes nonlinear) standardized color space built for the digital world

CIE-Lab

(L stands for luminance;
a controls green/red, b controls blue/yellow)



Linear RGB



CIE-Lab

*Nonlinear function of CIE-XYZ (RGB),
but is a perceptually uniform space*

"RGB to XYZ: The Science and History of Color" by John Austin

Additive vs. Subtractive Color

Additive

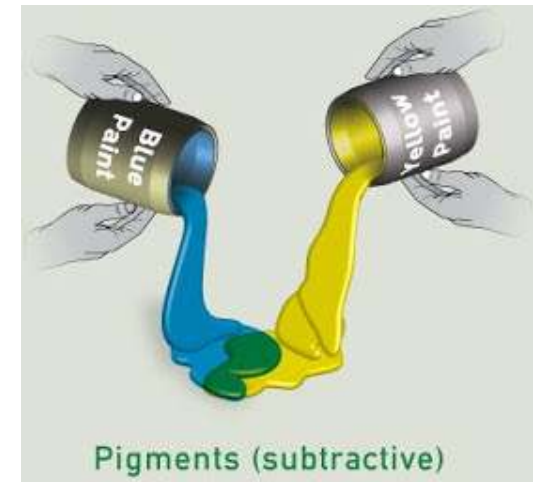
The resulting spectrum is the sum of the spectrums you are shining



Paints mixing or white light moving through successive filters

Subtractive

The resulting spectrum is filtered by each reflectance function



Paints mixing or white light moving through successive filters

Additive vs. Subtractive Color

Subtractive

The resulting spectrum is the sum of the spectrums you are shining

The resulting spectrum is filtered by each reflectance function

Are the red green and blue primaries better for additive or subtractive color mixing?



additive

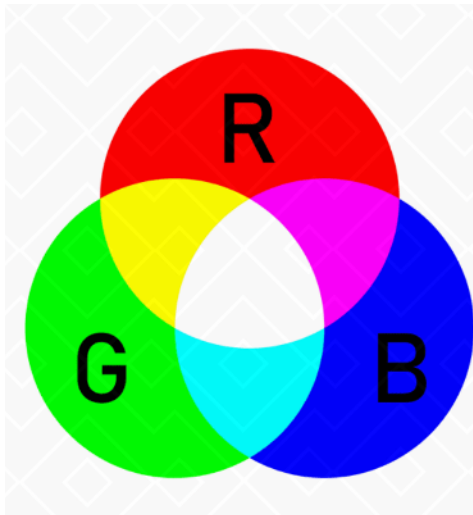


subtractive

Additive vs. Subtractive Color

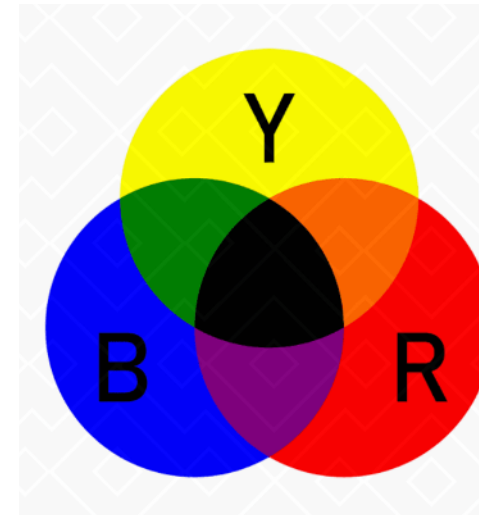
Additive

The resulting spectrum is the sum of the spectrums you are shining



Subtractive

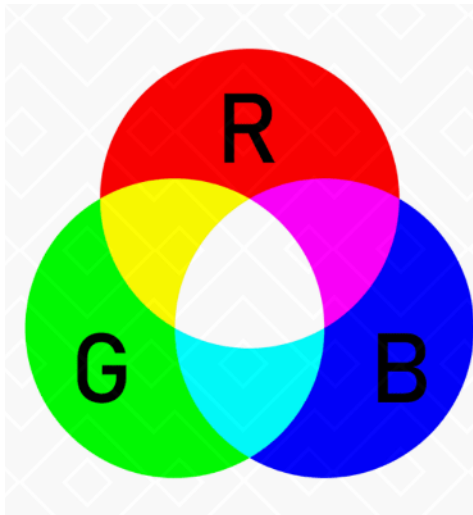
The resulting spectrum is filtered by each reflectance function



Additive vs. Subtractive Color

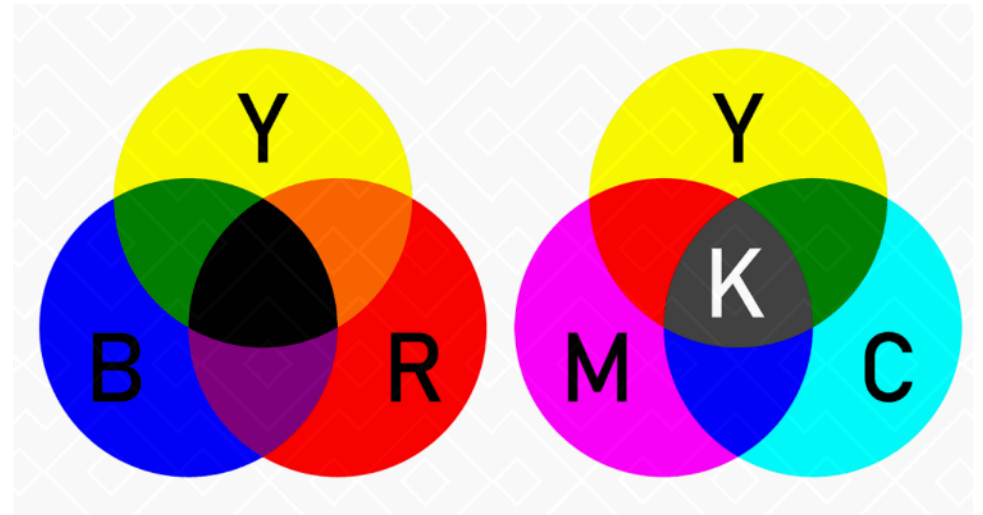
Additive

The resulting spectrum is the sum of the spectrums you are shining

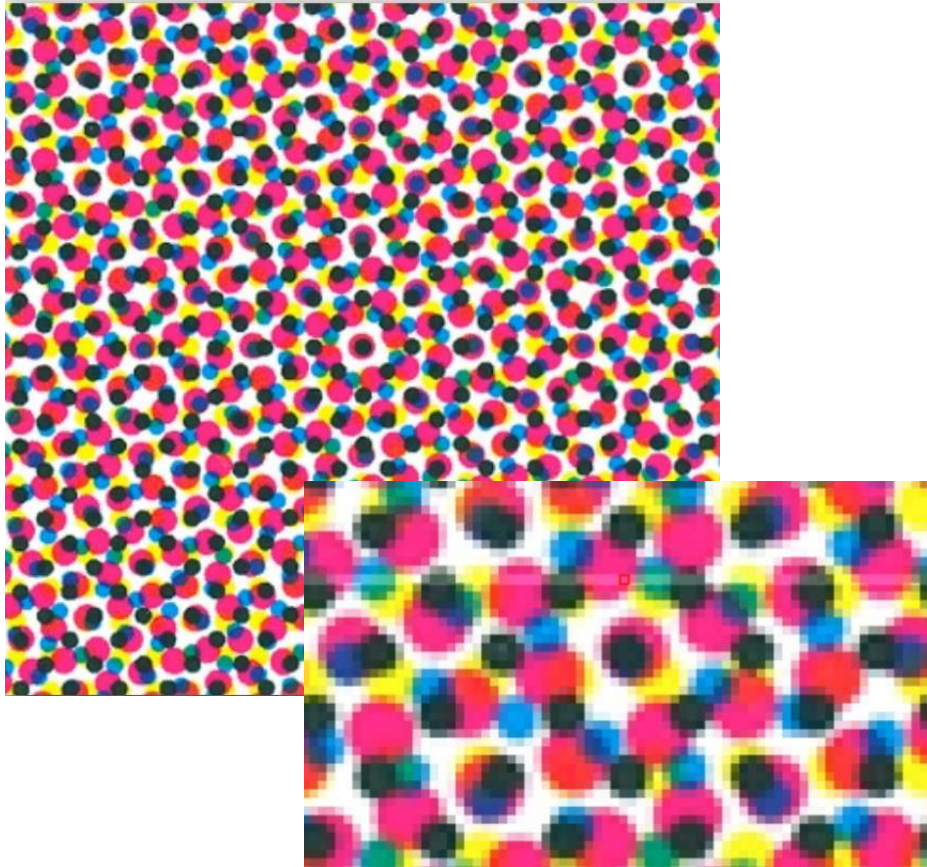


Subtractive

The resulting spectrum is filtered by each reflectance function



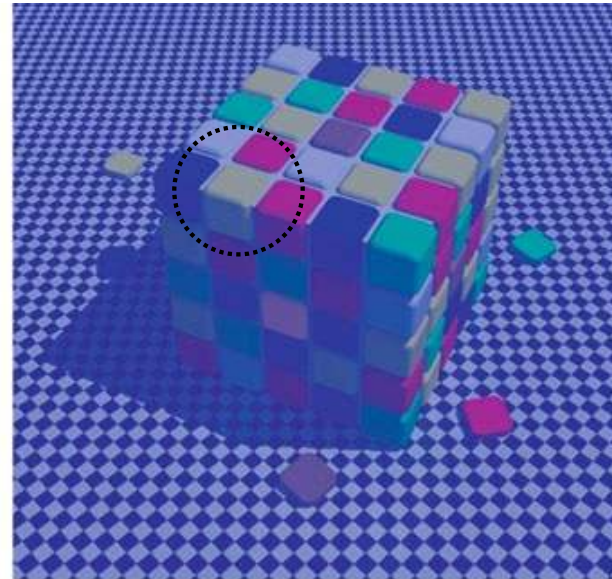
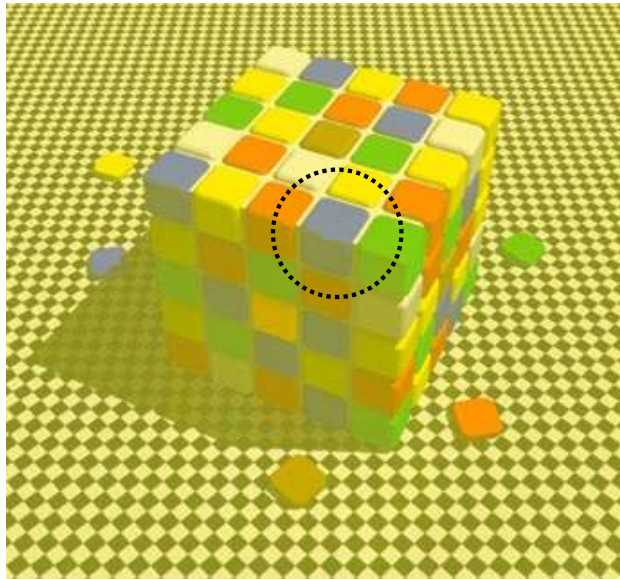
Color Printing



patches of the 3 subtractive primaries (C,M,Y) overlap partially on the page, making patches of 8 meta-primaries (Wh,C,M,Y,R,G,B,K), which combine additively in the eye when viewed from a distance

Even with all this...color is subjective

Retinal Response $\neq$ What Our Brain Sees



Retinal vs perceived color



Retinal vs perceived color

Important for “white balancing”

Retinal vs
perceived color.



Everybody sees colors just a little differently



- ✓ white/gold
- ✗ Blue/black

Questions?